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REGULAR CONTRIBUTION

A systematic literature review for network intrusion detection system (IDS)

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Abstract
With the recent increase in internet usage, the number of important, sensitive, confidential individual and corporate data passing through internet has increasingly grown. With gaps in the security systems, attackers have attempted to intrude the network, thereby gaining access to essential and confidential information, which may cause harm to the operation of the systems, and also affect the confidentiality of the data. To counter these possible attacks, intrusion detection systems (IDSs), which is an essential branch of cybersecurity, were employed to monitor and analyze network traffic thereby detects and reports malicious activities. A large number of review papers have covered different approaches for intrusion detection in networks, most of which follow a non-systematic approach, merely made a comparison of the existing techniques without reflecting an in-depth analytical synthesis of the methodologies and performances of the approaches to give a complete understanding of the state of IDS. Nonetheless, many of these reviews investigated more about the anomaly-based IDS with more emphasis on deep-learning models, while signature, hybrid-based (signature + anomaly-based) have received minimal focus. Hence, by adhering to the principles of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), this work reviewed existing contributions on anomaly-, signature-, and hybrid-based approaches to provide a comprehensive overview of network IDS's state of the art. The articles were retrieved from seven databases (ScienceDirect, SpringerNature, IEEE, MDPI, Hindawi, PeerJ, and Taylor & Francis) which cut across various reputable journals and conference Proceedings. Among the 776 pieces of the literature identified, 71 were selected for analysis and synthesis to answer the research questions. Based on the research findings, we identified unexplored study areas and unresolved research challenges. In order to create a better IDS model, we conclude by presenting promising, high-impact future research areas.

Keywords Network intrusion detection system · Signature-based · Anomaly-based · Artificial intelligence · Rule-based · Pattern matching

1 Introduction

The amount of important, sensitive, and confidential personal and business data traveling across the global network has dramatically increased in light of the significant rise in internet usage. Attackers have made attempts to breach the network's security through flaws in the security measures in order to have access to sensitive data, which could affect the operation of the systems and change the data's availability and confidentiality. Thus, cybersecurity has come to be seen as an essential field which proffer critical measure to combat cyberattacks and reduce associated costs and damages. To this aim, the intrusion detection system (IDS), which is a vital component of the security architecture, acts as a tool to recognize various forms of intrusions. IDS monitors a network and identifies suspicious and malicious activities or policy violations through network traffic, which enables network administrators to keep a constant watch on current threats [1].

IDS is classified into host-based and network-based depending on the environment and placement [2]. Host-based IDS is deployed on a single host or device to monitor data packets from the device by exclusively scanning through for

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suspicious actions and security policy breaches. The main drawback of this approach is the requirement to install an IDS on each host that needs to be guarded. Furthermore, a host-based IDS is probably going to require more processing time for each host, which will ultimately reduce its performance. In contrast, network-based IDS is deployed on the entire computer network to monitor, capture and analyze traffic to detect malicious activities. IDSs use two primary approaches for attack detection, which are either anomaly-based or signature-based [3]. An IDS works by searching for either signatures of known attacks or deviation from a predefined profile of normal activity. Typically, the signature-based IDS, also referred to as "misuse intrusion detection" or "knowledge-based intrusion detection," is reactive and presents the attacks in the form of signatures and patterns, which are then maintained as a database of attacks. These signatures/patterns are then compared with the data received on the network; if there is a match, the system flags such activity as malicious [4]. This approach is easy to develop and has the advantage of easily identifying known attacks with very low false positives. However, it lacks the capability of detecting novel/zero-day types of attacks; this is because it can only detect those attacks that are in its database. Also, it consumes enormous resources due to the extensive signature database that has to be sustained, updated, and compared with the data packets for possible attacks. On the contrary, anomaly-based intrusion detection systems, often known as "behavior-based IDS," creates a model of the normal behavior of the system, then look for activities that deviates from the model to identify attacks based on irregularities in the pattern relative to the created normal pattern [5]. The strength of this approach lies in its ability to detect new unknown attacks; however, it typically leads to a higher false-positive rate, which limits its application.

Researchers have made many attempts to come up with IDS methods capable of detecting attacks efficiently. Few are based on the signature-based approach using rule-based and pattern matching techniques, while recently, most of IDSs presented in the literature have focused on the anomaly-based approach using artificial intelligence (deep learning and machine learning techniques) to identify intrusions in network [6]. In contrast, minimal effort has been drawn toward exploring a hybrid-based technique that uses both signature and anomaly approaches.

Nonetheless, despite the fact that many studies have been conducted in the context of IDS, the current IDS methods still have a high false alert rate [7–11] and minimal detection rate [10, 12–14], and as a result, there has been a shift towards deep learning methods, which have been used by researchers to mitigate some of these issues. However, despite this effort, these approaches also suffer from computational and resource constraints, false-positive rate, and the problem of class imbalance in data [8, 10, 15–17].

Numerous review papers have examined various network intrusion detection methodologies, the majority of which take a non-systematic approach and merely compare the currently used approaches without reflecting an in-depth analytical synthesis of the methodologies and results of the approaches to provide a thorough picture of the IDS state of the art. Also, a lot of these studies focused more on the anomaly IDS with deep learning models, while signature- and hybrid-based (signature + anomaly-based) had a much narrower focus. Unlike earlier review studies, this work explores the current methods for signature-, anomaly-, and hybrid-based IDS, gives a thorough analysis of the most recent methods, datasets, attack types, and performances, and discusses the benefits, drawbacks, and perspectives. The novelty of this review paper lies in its effective in-depth synthesis and analysis of the various solutions present, giving more indications for wise choices, with a summary of the pros and cons of the various categories of algorithms, and a vision of the future evolution of these algorithms, additionally, previous review papers failed to review hybrid-based (signature + anomaly-based) IDS, this gap has been filled by this current study.

The remainder of the paper is structured as follows: Section 2 discusses related works. Section 3 discusses the study's methodology, which includes the review protocol, research questions, search strategy, selection of records, inclusion and exclusion criteria, quality assessment, data extraction. The study selection, publishing distributions, and discussion of results and findings are all included in Section 4's. Section 5 outlines the summary of discussions, in Sect. 6, challenges and future directions were discussed, while conclusions and viewpoints are drawn in Sect. 7

2 Related work

A thorough evaluation was carried out by [18] on IDS with a focus on the signature- and anomaly-based approaches; however, a systematic literature review approach was not followed, and there was also no in-depth analysis or comparisons of the methods, datasets, attacks, or performances of the techniques. In the same vein, [19] followed a systematic approach to reviewing hybrid-based IDS; however, the hybrid-based in this study emphasized hybridization by combining techniques rather than approaches; also, there was no synthesis or comparison of the performances of the techniques. Similarly, [20] presented a systemic evaluation of the literature on intrusion detection systems with a focus on the IoT environment; the hybrid approach was not considered; also no record of analysis and synthesis of the methodologies was found. In another study by [21], intrusion detection was reviewed in the network environment. The study was conducted following a traditional approach, which only shows

Table 1 Comparison with related reviewed papers

Paper	Year	Systematic study	Signature-based	Anomaly-based	Hybrid-based
[18]	2021	x	√	√	x
[21]	2021	x	x	√	x
[22]	2021	√	x	√	x
[19]	2022	√	√	√	x
[20]	2022	√	√	√	x
[23]	2022	√	x	√	x
[140]	2020	x	x	√	x
[141]	2013	√	x	√	x
Current study	2022	√	√	√	√

Key: √ Yes x No

a mere comparison of techniques without detailed synthesis or analysis. Finally, a systematic review was carried out by [22, 23] on anomaly-based IDS, which only focused on deep-learning techniques; also, no detail was provided on signature- or hybrid-based IDS. Therefore, in order to present the current state of the art in signature-based, anomaly-based, and hybrid-based IDS, this study is being done to cover the gaps in these evaluated articles. The comparison of the current review papers is shown in Table 1.

3 Research methodology

The study followed a systematic literature review (SLR) approach in identifying, evaluating, and interpreting all the relevant parts of the research studies that address specific objectives. This SLR adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses standards (PRISMA) [24] as well as the standards established for conducting systematic reviews in the field of software engineering [25, 26]. In carrying out the SLR, the study followed the following systematic procedures;

3.1 Literature review planning protocol

For this review, the following review planning protocol was taken into account: review background definition, determination of the need for SLR, identification of research questions, search strategy, Quality Assessment, data extraction, selection criteria for studies, and data syntheses.

3.2 Research questions (RQ)

- What methods or techniques are used for signature-, anomaly-, and hybrid-based IDS?
- What datasets are used for evaluating the signature-, anomaly-, and hybrid-based IDS?
- What are the performance metrics used in evaluating signature-, anomaly-, and hybrid-based IDS?
- What types of attacks are detected in signature-, anomaly-, and hybrid-based IDS?
- What is the performance level of the techniques employed in signature-, anomaly-, and hybrid-based IDS studies?
- What are the challenges in Signature, Anomaly, and Hybrid-Based IDS?

3.3 Search strategy

This segment shows how the literature on signature-, anomaly-, and hybrid-based IDS was searched and found. Using precise keywords that were developed in answer to the study questions, an automatic search technique was employed to explore and obtain the relevant academic papers from the seven (7) literature databases. The keywords used to build the search strings were chosen based on terms that are often used in literature and terms that are relevant to this investigation (i.e., signature-based methods applied to network IDS, anomaly-based methods applied to network IDS, hybrid-based approach applied to network IDS). For the execution of the SLR, specific keyword strings were mapped out and applied for each database: ("Network Intrusion Detection System" OR "NIDS" OR "Intrusion Detection System" OR "IDS") AND ("Signature-Based" or "Misuse Detection"), ("Network Intrusion Detection System" OR "NIDS" OR "Intrusion Detection System" OR "IDS") AND ("Anomaly-Based" or "Anomaly"), ("Network Intrusion Detection System" OR "NIDS" OR "Intrusion Detection System" OR "Network Intrusion Detection System" OR "NIDS" OR "IDS") AND ("Machine Learning" or "Deep Learning" or "Artificial Intelligence"), ("Network Intrusion Detection System" OR "NIDS" OR "Intrusion Detection System" OR "IDS") AND ("Rule-Based" or "Pattern Matching")

Literature for anomaly covers from 2018 to 2022, while signature-based and hybrid-based was from 2014 to 2022. Filters were not applied for the publication type or language while retrieving the records from the databases. First, keywords for discovering surveys, systematic reviews, and primary investigations were established. For searching primary research and survey publications in the context of the signature-based network IDS, the following search strings were employed:

- Signature-based intrusion detection survey
- Signature-based intrusion detection review
- Signature-based network intrusion detection system
- Intrusion detection system for misuse or signature detection
- Network intrusion detection system overview
- Systematic literature review on signature-based IDS

In the anomaly-based network IDS context, we employed the following search strings:

- Anomaly-based intrusion detection survey
- Anomaly-based intrusion detection review
- Anomaly-based network intrusion detection system
- Intrusion detection system for anomaly detection
- Network intrusion detection system overview
- Systematic literature review on anomaly-based IDS
- Intrusion detection system using machine learning and deep learning

For searching survey articles and primary studies in the hybrid-based network IDS context, the following search strings were employed:

- Anomaly- and signature-based intrusion detection review
- Literature review on intrusion detection system using the hybrid of signature- and anomaly-based detection
- Misuse and anomaly network intrusion detection
- Network intrusion detection using a hybrid approach of signature and anomaly

3.4 Selection of records

Duplicate copies were eliminated during the primary records selection process, and the records that remained were then screened by title and abstract to omit lecture notes, reports, books, and other sources outside journals and conference proceedings. This restricted the range of works that might be published in credible journals and proceedings. By using eligibility (inclusion and exclusion) criteria created in answer to the research questions, primary records were filtered to create secondary records. These standards were used to make sure that only pertinent studies were included in the data analysis.

3.5 Inclusion criteria

- Articles related to network intrusion detection system
- Relevant content to signature-based, anomaly-based, hybrid-based IDS
- Publication of articles in peer-reviewed journals.
- Accessible research articles.

3.6 Exclusion criteria

- Research articles dated before the year 2014 for the signature- and hybrid-based
- Research articles dated before the year 2018 for the anomaly-based
- Research articles not related to network-based intrusion detection system
- Research articles published in predatory journals
- Inaccessible articles.
- Irrelevant to anomaly-based and signature-based intrusion detection system
- Research articles that do not present experimentation or comparison results

3.7 Quality assessment (QA) of the selected eligible records

The secondary selected articles underwent a quality assessment test to show the significance of the reviewed papers to the work's contribution [26]. It is vital to adhere to a few stages by using a quality evaluation tool in order to examine the caliber of diverse investigations. The best articles for the study were chosen based on examination using a set of five QA questions, which are as follows:

- Does the paper's topic connect to network-based IDS?
- Is there a clear explanation of the backdrop issue the study has addressed?
- Is the research approach properly described and interpreted?
- Does the publication include the findings of its experiments?
- Do the paper prospects to perform additional future work as perspectives?

Each QA question is attributed a score of 1, implying that papers that fulfill all five (5) QA criteria have a total score of five (5). In scoring each paper based on the five (5) defined

criteria, there are two determining factors for the paper to be considered a quality paper and considered for the study:

- The paper must score a total point of 3 and above
- The paper must fulfill criteria QA1, QA3, QA4.

3.8 Data extraction and synthesis of the reviewed papers

The survey was performed on April 18, 2022. The literature search was conducted on seven (7) widely recognized databases with scientific scope; these are ScienceDirect, SpringerNature, IEEE, MDPI, Hindawi, PeerJ, and Taylor & Francis. A total of 776 papers were searched, out of which 705 papers were rejected and excluded using the exclusion and quality assessment criteria, while 71 papers having passed the quality assessment were selected for this review. Significant data were extracted from each of the selected studies. The extracted data included the authors of each paper, publication date, methodology, datasets, performance metrics (accuracy, F1-measure, precision, false alarm rate (FAR), recall, detection rate (DR), true-negative rate (TNR), true-positive rate (TPR), and false-negative rate (FNR), processing time (PT), throughput (TP), and memory usage (MU)), and limitations of the studies. By highlighting the advantages and disadvantages of the various strategies, the data were then aggregated for an in-depth investigation of the problems with intrusion detection in networks. Below are the data extraction fields:

- Employed methods or techniques
- Datasets that have been employed in the evaluation of the system
- The desired (output) predictions (performance metrics)
- Strength
- Limitations

4 Analysis and result

4.1 Studies selection and quality assessment

The literature database (n=767) and additional sources (n=9) yielded a total of 776 records; duplicates (n=216) were then eliminated, and 244 records (books, lectures note, and miscellaneous) were discarded out of the remaining 560 records that were put through primary selection. As a result, 316 records were determined to be journal articles and conference proceedings (287 journal articles and 29 conference proceedings); 78 records were therefore removed (reviews, surveys, and reports). Finally, after applying eligibility (inclusion and exclusion) criteria to 238 records, 97 studies were determined to be eligible. Only 71 studies, however, met the quality evaluation standards. To be analyzed and synthesized were these 71 studies. Fig.1 depicts the flowchart for the selection of the studies.

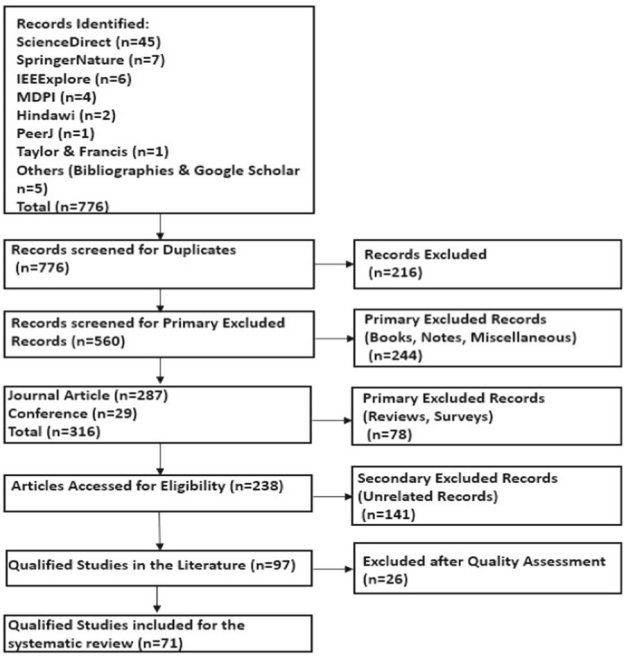

Fig. 1 Study selection flowchart

4.2 Distribution of publications among online database, journals and conferences

The 71 articles that were chosen for this study were published in a variety of publications and conferences, with 11 of them appearing in conferences and 60 in journals. The majority of the journals reviewed are in the network and security space; of the conferences, Materials Today Proceeding leads with three publications, followed by Procedia Computer Science with two publications, and the remaining conferences each have one (Table 2). The conference papers include a wide range of subjects, such as engineering, electronics, and data science. The distribution of publications by year and IDS's methodology is shown in Figs. 2 and 3. A trend toward anomaly-based network intrusion detection is clearly present. This number reflects an increase in interest in this field, particularly after 2018. In comparison, the hybrid-based have a minimal trend. Most of the articles reviewed were published in the year 2020, followed by 2021, and then the year 2019. The publication distribution among databases, journals, and conferences is depicted in Tables 2 and 3. The distribution of publications by year and IDS's approach is shown in Figs. 2 and 3.

Table 2 Publication distribution of journals and conferences, databases

Title of journals	No
Journal of Computer & Security	11
Journal of Computer Networks	5
Journal of Computer Communications	3
ICT Express	3
Knowledge-Based Systems	3
Advancing Technology for Humanity	5
Expert Systems with Applications	3
Computer & Electrical Engineering	2
Neural Computing and Applications	1
Future Generation Computer Systems	1
Engineering Application of Artificial Intelligence	1
Machine Learning with Application	1
Scientific African	1
Applied Intelligence	1
Cluster Computing	1
Cyber Security	1
Microprocessor & Microsystems	1
Journal of Visual Communication & Image Representation	1
Computer Networks and Communication	1
Security & Communication Networks	1
Journal of Network and Computer Application	1
Journal of Computational Science	1
Journal of King Saud Uni. Computer & Info. Science	1
International Journal of Computing & Digital System	1
Sensors	1
Evolutionary Intelligence	1
Electronics	1
Mathematics	1
PeerJ Computer Science	1
Journal of Information Privacy and Security	1
Soft Computing in Data Analytics	1
Preprints	1
IJITEE	1
Total	60
Titles of conferences	**No**
Materials Today: Proceeding	3
Procedia Computer science	2
Frontiers and Advances in Data Science	1
Global Transition Proceedings	1
Artificial Intelligence and Evolutionary Computations in Engineering System	1
Computer Science and Electronic Engineering Conference	1

Table 2 (continued)

Titles of conferences	No
Advances in Intelligent Info. Hiding & Multimedia Signal Processing	1
International Conference on Information Technology	1
Total	11

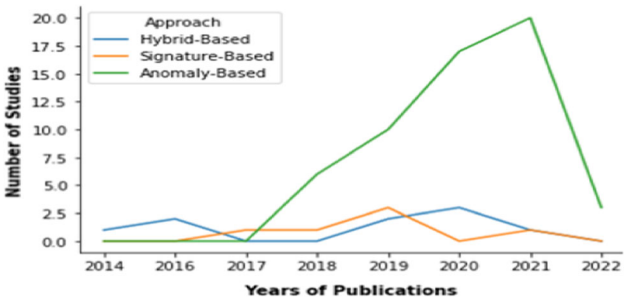

Fig. 2 Distribution of studies by type of ids approaches

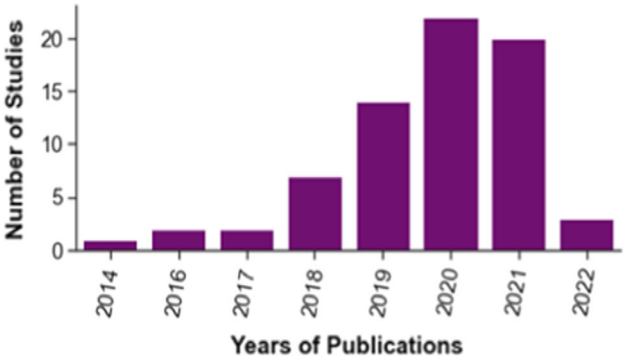

Fig. 3 Distribution of selected studies by year

Table 3 Publication distribution among databases

Online databases (publishers)	Publication
Science Direct	45
Springer Nature	7
IEEExplore	6
MDPI	4
Hindawi	2
PeerJ	1
Taylor & Francis	1
Others	5
Total	71

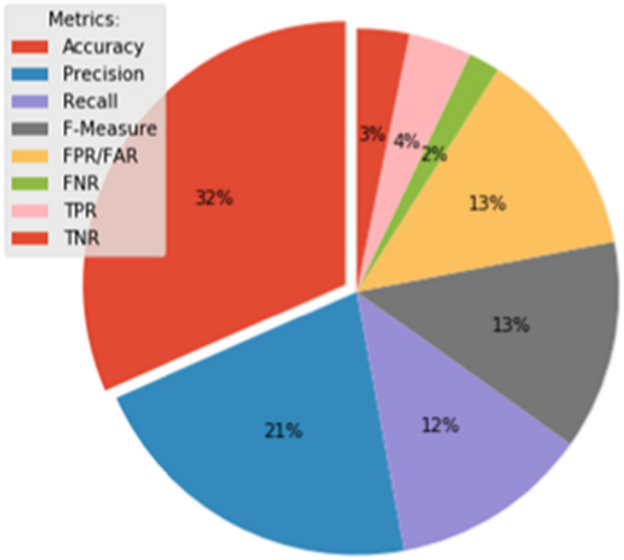

Fig. 4 Frequency distribution of performance metrics

4.3 Result distribution of anomaly-based intrusion detection system

The most employed metrics in assessment of anomaly-based IDS include accuracy, recall, precision, false-positive rate (FPR), also regarded as false alarm rate (FAR), f- measure, false-negative rate (FNR), and true-positive rate (TPR) (Fig. 4). The result also depicts that in terms of performance, [142] report that accuracy, precision, and recall are nearly 100%, with a false alarm rate of close to zero based on CNN and AE. By utilizing both linear and nonlinear dimensionality reduction, AE methods can identify anomalies. AE also can deal with high-dimensional, unstructured, and heterogeneous data generated from network traffic. İn addition, the training phase of AE involves the regeneration of new input data from partially destroyed ones. Automatic detection of important features, as well as independent learning of the key features of each class, is one of the attributes of CNN [15, 27–30], which is why it is considered a more attractive technique.

Nonetheless, factors like kernel size, padding, number of filters, and stride may affect the efficiency of CNN. However, combining AE with CNN will likely reduce data dimensionality and yield promising performance results. Nonetheless, the training and processing phase of the model may take too much time, resulting in high computational complexity and resources consumption; in addition, the model focused on a few types of attacks.

A model for IDS was developed in a related work [15] by using the CNN technique on UNSW-NB15 dataset, and the model had a 94.4% accuracy rate. However, the model did not address the class imbalance problem in data which caused a significant model downgrade; also, the detection rate was significantly low and could not effectively detect a zero-day attack. Similar results were obtained by [31] when they applied ANN to the CSE-CIC- IDS2018 dataset. They achieved 99.9% accuracy, recall, and precision while having a low false alarm rate. However, the model only considers a few different attack types, and training and preprocessing take an excessive amount of time.

Swarna Priya [32] suggested a hybrid strategy using PCA-GWO and AE classifier on the Kaggle dataset. The proposed model attained a 99.9% accuracy rate, a 95.4% sensitivity rate, and a 100% specificity rate. However, the model is resource demanding and multiclass classification was not considered. An information entropy deep believe networks (IE-DBN) model was proposed [33] for network IDS using the KDDCup99 dataset. Information gain (IG) was employed to minimize their dimensionality and delete any unnecessary features, while the number of hidden neurons in the DBN network and its depth were determined using information entropy (IE). To address the issue of data class imbalance, the synthetic minority oversampling technique (SMOTE) algorithm was also employed. 98.76% detection accuracy and 0.76 false alarm rate were obtained, which shows that using IE and IG with DBN improves the model's learning efficiency and detection accuracy, while the computational cost of the network and FAR are reduced. However, the class imbalance in data can be improved as the study has shown lower detection accuracy towards the minority class of attacks, with regard to larger datasets, even by using SMOTE oversampling technique. Furthermore, the researchers haven't thought about testing their model on contemporary attack types or using measures other accuracy and false alert rate.

Similarly, [30] proposed a model that achieved 99% accuracy, using a CNN technique on the CSE-CIC-IDS2018 dataset. However, the method focused only on denial-of-service (Dos) attacks and was resource-consuming.

A Wrapper-Based Genetic Algorithm and Naive Bayes (GANBADM) Model were proposed by [34] for detection of anomalies using NSL-KDD datasets. The wrapper-based genetic algorithm was applied for dimensionality reduction, while classification was done on the reduced dataset using naive Bayes. Results revealed that the proposed model had a 0.6% false alarm rate and an overall accuracy of 99.73% outperforming other classifiers like random forest, SVM, and decision tree. However, it has a low f-score and longer execution time.

The artificial bee colony (ABC) and artificial fish swarm (AFS) algorithms are combined in [35] to propose a unique hybrid classification approach. On the NSL-KDD and UNSW-NB15 datasets, the model was evaluated. The correlation-based feature selection (CFS) and fuzzy C-means clustering (FCM) techniques were used to separate the training dataset while removing irrelevant features. Furthermore, based on the selected features, CART technique was used

to generate an if-then rules, which was used to train the model and distinguish the anomaly and normal records. Results obtained demonstrated that the suggested method had a 99.9% accuracy rate, a 0.01% false alert rate, and a 99.99% detection rate. The overhead of the suggested method is comparable to that of the existing related methodology, according to examination of computational complexity and time costs.

In order to automatically choose the optimum feature subset and hyperparameters during the pre-training phase, [36] presented a twofold particle swarm optimization (PSO) technique. On the CICIDS2017 and NSL-KDD datasets, the model was assessed, and its performance differences were assessed using three distinct deep learning techniques: deep neural networks (DNN), long short-term memory recurrent neural networks (LSTM-RNN), and deep belief networks (DBN). The analysis of the result shows that the proposed model yielded a notable improvement in detecting intrusions in network. The proposed model achieved a significant increase from 4 to 6% detection rate, and a reduction rate from 1 to 5% false alarm rate compared to the comparable values of the same models without pre-training on the same dataset. The double PSO combined with DBN achieved the best detection rate of 99.81% and 0.23% false alarm rate on the NSLKDD dataset, while on the CICIDS2017 dataset, a detection rate of 99.92% and 0.1 false alarm rate was achieved.

On the KDDcup99 dataset, [17] presented a novel model for intrusion detection using ensemble of discriminant classifiers. Testing and training were conducted using random subspace algorithm. The result demonstrated that in comparison with previous method, the model achieved better detection accuracy in detecting all types of attacks classes for DOS, Probing, R2L and U2R with an overall accuracy of 98.9%.

The performance of the IDS classifier was improved by [37] by choosing relevant features for each category of attacks rather than attributes that were common to all attacks. This model was based on the genetic algorithm and multiple SVM classifiers. On the basis of the CICIDS2017 and ADFA-LD datasets, the performance of the suggested system was assessed. The model, which had a 99% accuracy rate, shows the possibility of using GA-based feature selection for IDS to identify attacks accurately while requiring less computing complexity and communication overhead.

The Mutual Information Firefly algorithm (MIFA) was employed in the wrapper for feature selection in the model for IDS suggested by [14], while Bayesian network (BN) and C4.5 were used as the classifier. On the KDDCUP99 dataset, the model was evaluated. According to the experimental findings, ten features can improve intrusion detection's precision. Comparing the suggested work to the existing work reveals promising advancements. KDD CUP 99-based experiments revealed that the proposed work improved both accuracy and false alarm rates. The model achieved an accuracy of 99.98%, 63.85%, 98.73%, 17.24% for C4.5 and 99.95%, 93.42%, 97.83%, 68.97% for BN, respectively. A false-positive rate of 0.01 for DoS, 0.04 for Probe, 0.04 for R2L, and 0.04 for U2R attacks, respectively, was also attained. The proposed paradigm, however, requires lengthy computation.

SVM ensemble with feature augmentation was used by [7] to achieve intrusion detection. The original features were initially transformed using the logarithm marginal density ratios in order to particularly build a new and superior altered training set. The intrusion detection model was then created using the SVM ensemble. The experiments' findings demonstrated that the proposed model had performance accuracy of 99.91%, detection rate of 99.92%, and false-positive rate of 0.11%.

Zhang et al. [10] developed a comprehensive model that integrates long short-term memory and multiscale convolutional neural network (MSCNN-LSTM). The spatial and temporal aspects of the dataset were analyzed by MSCNN and processed by LSTM, respectively. In order to achieve the classification, the model used the spatial--temporal features. On the UNSWNB15 dataset, the system was evaluated. The MSCNN-LSTM model outperforms conventional neural networks, according to experimental findings, having a better accuracy of 95.6%, false negative rate of 1.6%, and a false alarm rate of 9.8%. However, it exhibits high computational time when compared with others. Also, the model did not consider rare attacks, neither did it tried to address the problem of imbalance datasets.

Li et al. [38] proposed the AE-IDS lightweight online approach for IDS (autoencoder intrusion detection system). The model was trained using normal traffic rather than a labelled dataset. The method applied random forest for effective selection of the most significant features, then the chosen features were classified into multiple subsets using the affinity propagation (AP) algorithm to achieve feature grouping, this was carried out to identify the features with high correlation that can enhance AE expression ability. Finally, AE, KMeans, and Gaussian mixture model (GMM) were applied for anomaly detection in the network traffic; AE handles the incoming traffic and then calculates the root-mean-square error (RMSE) whose value is the criterion to give an insight as to whether or not the network traffic is normal. Class imbalance problem was also addressed using AE. Experimental results showed that the online-based AE-IDS was able to reduce computational complexity through feature selection and feature grouping. Also, in comparison with other conventional machine learning approach and some offline methods, it shows a better performance over them. However, due to the current nature of network attacks which has become more encrypted and as a result mixed with the normal network traffic, therefore extracting features from the network traffic is not sufficient to guarantee a good IDS model, therefore, it

is imperative to consider other information like system logs in order to have an IDS that can effectively defend against these malicious activities. In addition, the model was only evaluated using recall and computational time only.

A SVM and naive Bayes IDS framework were presented by [16]. In order to create new, high-quality data, the original features were modified using the naive Bayes technique, and the SVM classifier was trained using the altered data to create the intrusion detection model. Experimental findings showed a good performance accuracy of 93.75%, 98.92%, 99.35%, and 98.58% for UNSW-NB15, CICIDS2017, NSL-KDD, and Kyoto 2006+ dataset, respectively. Additionally, the strategy outperformed other techniques in respect to detection rate and false alarm rate. Although only binary case was considered for intrusion detection, there is need for the model to be extended on scenarios with different attack types. Also, class imbalance was not properly addressed in the methodology.

A wrapper-based feature selection strategy, known as TS-RF, was proposed in a study by [8] that used tabu search (TS) as a search method and random forest (RF) as the learning algorithm for the IDS. The UNSW-NB15 dataset was employed to assess the proposed model. In comparison with the gain ratio, Chi-square, and Pearson correlation feature selection procedures that were previously used, the testing findings indicated an enhanced classification accuracy of 83.12 and a false-positive rate of 3.7%. However, the dataset's class imbalance issue was not addressed, which could have a negative influence on the classifier's accuracy and increase the rate of false positives and misclassifications.

Wang et al. [13] proposed an IDS model based on deep belief network Softmax (DBN-Softmax) and stacked denoising autoencoder extreme learning machine (SDAE-ELM). In the first phase, SDAE was applied to learn the features of the dataset which then served as an input to the ELM algorithm where the features were fine-tuned to produce the trained model. Finally, the SDAE-ELM model was applied on the testing data for intrusion detection. Each layer in the DBN phase is a restricted Boltzmann machine (RMM), which implies that the entire network is a stack of different RBMs. Layer by layer, an unsupervised training process was used to train the complete network, and then, the back propagation (BP) technique was used to categorize the data using the Softmax classifier in the DBN model to determine the type of intrusion and increase accuracy. To optimize SDAE and DBN models, ELM algorithm and Softmax classifier were both used. Experimental result showed that both models produced a good detection accuracy at both levels of binary and multiclass classification over the traditional machine learning models. However, even though the SDAE-ELM model shows effective ability, its detection ability on small dataset is poor. Also, the DBN-Softmax model takes too much time to train on large datasets and not realistic for real-time detection.

Liu et al. [11] proposed an IDS model by applying adaptive synthetic (ADASYN) oversampling technique and LightGBM ensemble. ADASYN was applied to address the problem of imbalance data. As well as applying LightGBM ensemble was used as classifier and addressed the problem of computational complexity associated with training the model and intrusion detection time. The model was assessed utilizing UNSW-NB15, NSL-KDD, and CICIDS2017 dataset. According to experimental findings, LightGBM achieved an accuracy of 83.98%, 89.79%, and 99.86%, respectively for the three datasets. Furthermore, after applying ADASYN oversampling, the detection rate of minority classes was improved, thereby achieving an overall detection accuracy of 85.89%, 92.57%, and 99.91%, respectively. However, the rate of false positive is relatively high; therefore, the model can be improved to address this problem.

Shahraki et al. [39] examined the efficacy of the binary intrusion detection classifiers Modest AdaBoost, Real Adaboost, and Gentle Adaboost. Results showed that when compared to Gentle AdaBoost and Real AdaBoost, the Modest AdaBoost classifier performed worse and had a greater error rate. While Gentle and Real AdaBoost both perform roughly 70% better than Modest AdaBoost in terms of error rates, this demonstrates their suitability for binary classification in the context of intrusion detection. Modest AdaBoost, however, outperforms them with respect to processing time by roughly 7%. Nevertheless, the model didn't take multi-class classification into account.

Alazzam et al. [40] presented a pigeon inspired optimizer (PIO) feature selection algorithm which was targeted at reducing the number of features required for building a robust IDS, at the same time retaining a good rate of accuracy, and low false positives. Numbers of features in UNSW-NB15, NSL-KDD, and KDDCup99 was significantly reduced by the proposed PIO method from 49 to 5, 41 to 5, and 41 to 7, respectively. Experimental result also showed that a high accuracy, and TPR was maintained, with reduction in training time.

Zhou et al. [41] presented an ensemble system that combined multiple modified adaptive boosting with area under the curve (M-AdaBoost-A)- based classifiers into an ensemble by utilizing tactics like particle swarm optimization. The system was designed to effectively detect network intrusions while addressing the problem of class imbalance. The model did not call for data preprocessing like feature selection or variable scaling, which prevented data information loss. The model was successful in achieving a false positive rate of 0.00 and an accuracy of 0.99%.

Nguyen and Kim [28] present a novel algorithm for network IDS. The GA with KNN fitness function was applied

to select an improved feature subset, while fuzzy C means clustering (FCM) was employed to improve these selected features. CNN was further used as an extractor to produce high quality deep feature subset which has the ability to learn the majority of attack character kinds through several layers. Finally, BG was applied as a classifier validated using 5-fold cross-validation. Result showed that GA, FCM, and CNN was able to produce a high-quality feature set, while the hybrid of CNN and BG significantly led to an improved detection performance of 98.2%, 0.5 false alarm rate and true positive of 95.4%. However, the proposed model consumes a lot of time which was as a result of the GA and CNN that was implemented for the selection of the relevant features Furthermore, reiteration of the fivefold cross-validation also took lots of time for the practical simulation. Additionally, when used in the real world of computer networks, there is also the possibility of data imbalances that skewed the results in favor of the dominant class.

Recurrent convolutional neural network (RC-NN) for intrusion detection was proposed by [42]. The CNN, LSTM, and ant lion optimization algorithms were incorporated in the model. Using the DARPA and CSE-CIC-IDS2018 datasets, the proposed strategy was assessed and compared to existing methods. The model achieved a reduced error rate of 0.0012 and an increased classification accuracy of 94%. As a result, this method can identify each attack more accurately than previous research.

To significantly enhance the network anomaly detection model's capacity for generalization, [29] combine ResNet with semantic reencoding. With its extensive semantic coding space, the suggested model addressed network traffic anomaly detection with a promising result. However, semantic re-encoding technology has only slightly improved traffic identification performance for network traffic with low semantic space.

Zhao et al. [43] developed an IDS model that makes use of multi-layer perceptron neural networks (MLP-NNs) and multi-objective genetic algorithms (MOGA) in an effort to address the issue of high false positive and poor detection rate in detecting intrusions. For the NSL-KDD dataset, the proposed method produced a detection accuracy of 97% and a false-positive rate of 2%. The approach's biggest drawback is that it takes a long time and necessitates iterative computations. A more realistic dataset should be taken into account to demonstrate the applicability of the strategy because it was also tested using tiny subsets of the dataset.

Rajesh Kanna and Santhi [44] presented a combined model of hierarchical multi-scale LSTM (HMLSTM) and optimized CNN (OCNN) for intrusion detection. The hyperparameters of CNN were tuned using lion swarm optimization (LSO) for the best configuration for learning spatial information. The HMLSTM collects the time features and learns the hierarchical relationships between the various features. This information was then utilized to further classify the network data using the unified model. Datasets from NSL-KDD, ISCX-IDS, and UNSWNB15 were used to assess the model. Experimental findings revealed that the suggested model outperformed alternative IDS techniques. The results demonstrated that the OCNN-HMLSTM model effectively detected numerous attacks and improved intrusion detection in all three datasets.

Sona and Sasirekha et al. [45] suggested an IDS model based on KSIDO-PABCE, which stands for Kulczynski similarity indexed dragonfly optimization-based polytomous adaptive base class ensemble. This model is intended to offer greater accuracy while requiring less time. KSIDO was employed to pick features. The initial population of dragonflies, or features, in the search space is produced via the optimization process. Kulczynski similarity index is used to assess each dragonfly's fitness following startup. The best features were picked based on fitness evaluation. The PABCE technique was used to discover various attack kinds with the chosen ideal attributes. The proposed method increased detection accuracy for different types of attacks with minimal time.

An ANN-based IDS model was proposed by [12]. The dataset UNSW-NB15 was used. The outcome revealed that the method achieved 76.96% accuracy rate. However, the system's accuracy is poor. A lightweight NIDS paradigm with two primary parallel-operating subsystems was proposed by [46]. Each subsystem was trained using the one-class support vector machine (OCSVM) algorithm. While the second subsystem was trained on attack packets, the first subsystem was taught on normal packets. Together, the outcomes of the two subsystems were used to provide a sound assessment of each packet that moves across the network. On the datasets KDDCUP-99, NSL-KDD, and UNSW-NB15, the suggested model was assessed. The findings demonstrated that the suggested NIDS model outperformed alternative approaches that were already in use in the literature.

Disha and Waheed [47] used a variety of ML models for the binary classification job of detecting intrusion in networks, including decision tree (DT), multilayer perceptron (MLP), AdaBoost, gradient boosting tree (GBT), gated recurrent unit (GRU), and long short-term memory (LSTM). Using the datasets from UNSW-NB 15 and network TON IoT, the models' performance was evaluated. Twenty significant features from the UNSW-NB15 dataset and 10 features from the network TON IoT dataset were chosen using the integrated feature selection technique known as Gini impurity-based weighted random forest (GIWRF). To help the learning algorithm grasp the class distribution, GIWRF used the Gini impurity as a splitting criterion of trees and adjusted the weights for two different classes of the unbalanced data. The experimental findings demonstrated

that DT performed better using the feature selection method than other ML models. However, this study did not take into account multiclass classification or temporal complexity analyses.

A genetic-based enhanced grey wolf optimization (GB-EGWO) improved intrusion detection algorithm was presented in [48]. The genetic crossover technique was utilized by GB-EGWO, and it involves repeating the process of feature selection until the best search agent (solution) is realized before deciding whether to move the position in order to find the best solution for identifying network intrusions. The model was evaluated on NSL-KDD dataset. The GB-EGWO algorithm demonstrated improved attack detection accuracy of 98.62%. However, it is important to reduce the amount of time required to identify the attack in the dataset.

A two-stage hybrid model for IDS was introduced in [49]. First, smoothed l1 regularization with a sparse autoencoder (SAE) was used. Smoothed l1 regularization was utilized to guarantee the autoencoder's sparsity because it teaches sparse representations of features. Second, the attacks were predicted and categorized using the deep neural network (DNN). On the UNSW-NB15 dataset, experimental findings reported that the presented technique outperformed other traditional techniques with an accuracy of 99.98% and detection rate of 99.99%. However, the method can be enhanced utilizing a variety of autoencoders to better represent feature learning and dimensionality reduction processes.

Sharma and Yadav [50] used recursive feature elimination (RFE) technique with a number of machine learning techniques (decision tree, support vector machine, and an ensemble classifier random forest in the form of discriminant analysis) to identify attacks. RFE was used to eliminate any redundant or unnecessary features from the KDD CUP99 dataset, and it was crucial in lowering the dataset's dimensionality. The results of the experiments demonstrated that the suggested approach produced good classification rates for all classes of attacks. Prior to feature selection, a comparison of the three classification methods revealed that random forest performed better than SVM. SVM outperformed random forest and decision tree after feature selection was used, nevertheless.

Halim et al. [51] suggested the GA-based feature selection (GbFS) technique to improve the precision of classifiers. With the use of the datasets CIRA-CIC-DOHBrw-2020, UNSW-NB15, and Bot-IoT, the improved GA-based feature selection approach was put to the test. GbFS, which has a 99.80% accuracy rate, performed more accurately when compared to traditional feature selection techniques. However, because of the numerous repeating rounds and the replication processes, the application of GA led to fundamentally slow functioning. Additionally, the suggested system uses three supervised learning classes in the prediction phase; hence, an unsupervised learning technique, such as clustering, can be applied to enable the machine to learn new types of attacks on its own.

Mazini et al. [52] suggested a hybrid strategy based on the artificial bee colony (ABC) and AdaBoost algorithms. ABC was used to choose the most pertinent features, and the AdaBoost algorithm was used to categorize the chosen features. The proposed method had a 98.9% accuracy when tested on the NSL-KDD and ISCXIDS2012 datasets. The proposed technique outperformed other IDSs on the same dataset, according to the testing results. However, the dataset used does not reflect contemporaries' attacks of todays network.

Using the UNSW-NB15 dataset, [53] constructed a two-stage anomaly-based IDS technique. To choose the most crucial features, recursive feature elimination and random forests were used. Support vector machine, gradient boost machine, and logistic regression were then used to accomplish a binary classification and find aberrant traffic. With SVM, an accuracy of 82.1% was attained. To enhance the accuracy of identifying attack types, the output of the SVM was fed into a number of polynomial classifiers. The effectiveness of naive Bayes, decision trees, and polynomial SVM in particular was assessed. The use of the two-stage hybrid classification led to an improved accuracy of 86.04%, according to the results.

Using the UNSW-NB15 dataset, [54] investigated five machine learning techniques, including random forest, decision tree, logistic regression, K-nearest neighbors, and artificial neural networks, to identify different types of network attacks. Random Forest, one of the classifiers, had the best accuracy, coming in at 89.29%. Additionally, there was evidence of further advancement in classification model accuracy after Synthetic Minority Oversampling Technique (SMOTE) was used to address the issue of class imbalance, with Random Forest classifier achieving the highest accuracy of 95.1% with 24 chosen features from the principal component analysis method. Additionally, class balancing exhibited no beneficial effects on LR and ANN classifiers, instead reducing their accuracy after handling minority classes.

To solve the issues of high false-negative and false-positive difficulties in IDS, [55] presented a hybrid paradigm. To enhance the model's accuracy, information gain and vote technique were employed to find the most relevant features, while naive Bayes, AdaBoostM1, REPTree, random tree, J48, meta pagging, and decision stump were applied as classification algorithms. On the NSL-KDD dataset, the model was assessed. The findings revealed that accuracy had improved and that the rate of false positives had decreased. By using the suggested technique on other datasets with different optimization strategies, this work can be enhanced further.

A feature selection method based on two-level ensembles of classifiers was developed in the work of [56]. The feature size of the training datasets was reduced using

the ant colony algorithm, particle swarm optimization, and genetic algorithm techniques. To select attributes that maximize classification performance, a reduced error pruning tree (REPT) was utilized. Then, a two-level collection of classifiers called rotation forest and bagging approaches were used. Using the datasets from NSL-KDD and UNSW-NB15, the proposed system was evaluated. The proposed model outperformed other techniques, according to the authors, with accuracy rates of 85.8% and 91.3% in the NSL-KDD dataset and UNSW-NB15 dataset, respectively. This work can be enhanced upon with new techniques that utilize fewer features while producing an improved accuracy.

For a better IDS model, [57] merged the ensemble of feature selection (EFS) and adaptive grasshopper optimization algorithm techniques (AGOA). According to the report, there was an increased rate detection of 99.23%, accuracy of 99.13%, and a low false alarm rate of 0.067.

Devan and Khare [58] reported an XGBoost-DNN model for IDS, in which the XGBoost technique was utilized for feature selection and the NSL-KDD dataset was used to train a deep neural network (DNN) to detect network intrusions. The model's results were contrasted with those of existing naive Bayes, logistic regression, and support vector machines. The outcome revealed that DNN outperformed the current models in terms of classification accuracy, with a score of 97%. In another study, [59] applied AE and DNN to detect unknown intrusions. The model showed a better performance with respect to lightweight and low resource consumption; however, it had low detection accuracy and huge training time.

Support vector machine (SVM) classifier and information gain ratio (IGR) were used to pick the special set of features from NSL-KDD datasets, and [60] presents an efficient anomalous IDS based on these two techniques. SVM was employed for training and intrusion detection. The usage of SVM with Information gain ratio increased accuracy to 96.24% and resulted in a reduced false alarm rate, according to experimental findings.

Using the UNSW-NB 15 dataset, [61] suggested an AdaBoost-based method for network intrusion detection. MLP and SVM were contrasted with the suggested model. The findings revealed that the suggested method attained an accuracy of 99.3% and further indicated that it is possible to identify various types of network intrusions. Similarly, Using the NSL-KDD dataset, [62] suggested an unique random neural network-based IDS and achieved an accuracy of 94.5%.

Gao et al. [63] introduced an adaptive ensemble learning model combining multiple DT with an adaptive voting mechanism. The suggested model was assessed using the NSL-KDD dataset. According to the outcome, the adaptive accuracy was between 84.23 and 85.2%. In the same vein, [64] introduced a new DL model for NIDS. The KDD-Cup99 dataset and the CNN algorithm were used to create the model. Dimensionality reduction of the data using PCA and autoencoder produced altered data that was given to the CNN input layer. The features of the network traffic were then extracted and examined using the CNN model. A total accuracy of 94.0% was attained, but U2R and R2L attack types had low detection rates, with 20.61 and 18.96% of attacks being detected, respectively.

A new IDS framework based on feature selection and ensemble learning methods was proposed by [65]. In order to select the best subset of features, CFS-BA used the correlation between the features. After that, voting approaches were utilized to combine the probability distribution of base learners for attack detection, while random forest (RF) and C4.5 were integrated for attack detection. The CIC-IDS-2017, AWID, and NSL-KDD datasets were used to test the model. Results showed that when compared to the other approaches, the proposed algorithm had a greater performance accuracy of 99.81%.

To address IDS's poor performance and high false-positive rates, [66] presented an enhanced krill swarm technique based on the linear nearest neighbor lasso step optimization (LNNLS-KH). According to experimental findings, the LNNLS-KH algorithm deleted unnecessary features from the NSL-KDD and CICIDS2017 datasets while selecting 7 and 10 meaningful features from each dataset, respectively. According to the provided results, the suggested strategy improved intrusion detection accuracy and also performed well in terms of convergence speed and optimal fitness iteration curve. It also had a low false positive rate.

Gupta et al. [27] introduced the LIO anomaly-based IDS model, which is based on the improved one-vs-one (I-OVO) technique and long short-term memory (LSTM) classifier. The LIO IDS is divided into two tiers. The LSTM classifier is used in the first layer to distinguish intrusions from regular network traffic, while random forest and bagging ensemble algorithms are used in the second layer to categorize the identified intrusions into distinct types of attacks. The model also made use of borderline SMOTE, random oversampling, and SVM SMOTE approaches to address class imbalance in the data. The NSL-KDD, CICIDS2017, and CIDDS-001datasets were employed to assess the model. On the NSL-KDD dataset, the model had an accuracy and detection rate of 87.0% and 91%, whereas on the CICIDS2017 dataset, it had an accuracy and detection rate of 96% and 99%. The outcomes showed that the suggested strategy accurately detects intrusions in a timely manner.

Chiche and Meshesha [67] suggested a novel strategy to enhance IDS performance. The model was developed using random forest as a classifier. The NSL-KDD dataset was used to apply the suggested approach, and the results revealed an accuracy of 99.91%. For misuse intrusion detection, [68] presented an extreme learning machine with a

hybrid kernel function (HKELM). By combining the gravitational search algorithm (GSA) and differential evolution algorithm, the parameters of HKELM were modified, which enhanced both its global and local optimization capabilities during attack prediction. Additionally, the feature extraction and dimensionality reduction were accomplished using the kernel principal component analysis (KPCA) algorithm. Finally, the KPCA-DEGSA-HKELM new IDS technique was discovered. On the KDDCup99 dataset, the UNSW-NB15 dataset, and the dataset from the Tennessee Eastman process, the proposed technique was tested. The experimental findings support the suggested method's excellent accuracy as well as its time-saving advantages. Table 4 shows the analysis distribution of the various works.

4.3.1 Analysis of accuracy range

The range of accuracy for each employed technique for anomaly detection is shown in Table 5. ANN spans a large range, from 76.9 to 99.9%; this technique was used four times in total. Furthermore, RF covers a broad range, starting at 83% and ending at 99.8%, while DBN covers from 87.2 to 98.1% and CNN covers a range of accuracy from 87 to 99%. Five times, SVM was used with an accuracy range of 96.2–99.9%. CNN and LSTM were used four times each. With CNN and AE, which were utilized only once, a 100% accuracy was attained. Section 4.3.3 explains the different methods used in anomaly-based IDS, with emphasis on their contributions, strength and limitations. The various techniques used in the investigations are displayed in Table 5.

4.3.2 Analysis of datasets and types of attacks detected

Table 6 demonstrates that NSL-KDD and KDDCup99 are the most often utilized datasets for anomaly-based IDS. These datasets, however, do not represent contemporary attack types. The datasets that have been employed in the literature are displayed in Table 6.

The types of attacks an IDS is capable of detecting is an essential factor in identifying an effective IDS. While some of the models in the study have achieved great accuracy in identifying a certain attack type, their studies did not address modern types of attacks. Therefore, there is a need for a model that can detect as many attacks including modern ones with higher detection rate. For instance, the model proposed by [34] was applied to the NSLKDD dataset. Attacks of the user to root (U2R), remote to local (R2L), denial of service (DoS), and probing kind were identified by this model. Similarly, studies by [14, 17, 27–29, 33, 41, 43, 46, 48, 64, 69] identified the same attack types while utilizing KDDCup99 and NSL-KDD datasets. Furthermore, in [8, 10, 11, 13, 15, 35, 44, 49, 68, 70], NSL-KDD and UNSW-NB15 increased the attack surface by identifying contemporary attacks like shellcode, fuzzer and worm, exploits, back-doors, analysis, generic, and reconnaissance. [31, 38] applied CSE-CIC-IDS2018 that tackles attacks such as Bruteforce, Heartbled, Bot, Web attack, DoS-Hulk, Infiltration, DoS-GoldenEye, DoS-lowHTTPTest, DDoS-LOIC-HTTP, DoSSlowloris, and DDoS-HOIC. A variety of attacks, including DoS/DDoS, Botnet, Web Attack, Brute Force, Infiltration, PortScan, web attack, Bot, grey hole, black hole, SQL injection, Benign, DoS Hulk, DosHulk, FTP-Potato, SSH-Potato, Dos Slowloris, Dos Slowhttptest, Heartbleed were utilized in the study by [27, 37] using the CICIDS 2017 dataset.

In the study conducted by [7, 9, 16, 32, 36, 39, 40, 42, 71–73], the authors only focused on binary classification, either an attack or not an attack. In the studies [7, 11, 29], the kinds of datasets used were not revealed. The techniques employed in the earlier investigations have proven their capacity to identify a variety of attack patterns, including the attacks mentioned above. However, there are differences in their ability to detect different attack types, particularly as a result of datasets containing unbalanced data. Therefore, further study is needed to both improve the rate at which other assaults are detected as indicated above as well as the ability to recognize recent attack types with greater accuracy.

4.3.3 Machine learning (ML) and deep learning (DL) techniques for anomaly-based IDS

IDS using an anomaly-based approach rely on ML and DL algorithms for the IDS's training and prediction phases. This presentation provides an overview of the various ML and DL methods used in anomaly-based network IDS. A quick summary of ML and DL approaches, their strength and limitations, and references to related research are provided in Table 7.

Naïve Bayes (NB) The Bayes' Theorem is used by NB, a straightforward and highly scalable probabilistic classifier [74], to estimate the likelihood that an event will occur based on prior observations of related events [75]. During the stages of training and categorization, it works effectively. NB takes for granted that each vector property is equally important and independent. Based on the prior data, NB is used in intrusion detection to predict the possibility that a class is either a normal class or an attack class. When deciding whether to categorize unlabeled traffic as normal or anomalous, NB estimates the posterior probability. To predict the likelihood that seen data will be normal or abnormal, a separate set of features from the observed traffic, such as status flags, protocol, and latency, are used. NB classifier has been used by several IDSs to identify abnormal traffic since it is a straightforward and easy to implement algorithm [16, 34, 71]. It can categorize using binary or multiple labels and only needs a

Table 4 Techniques, datasets, and performance of anomaly-based, studies models

Paper	Year	Classifier	Dataset	Metrics	Pros	Cons
[15]	2021	CNN	UNSW-NB15	AC: 95.6 P: 97.9	Shows significant improvement in multiclass model towards detecting novel attacks	Low detection rate for the underrepresented classes Significant model downgrade resulting from class imbalance problems
[31]	2019	ANN	CSECIC-DS2018	AC: 99.9 P:99.9 R: 99.9 FM: 99.9 FAR: 0.03	High detection rate, with low false positive	Huge training and processing time Detects only a few types of attacks
[34]	2021	NB + GA	NSL-KDD	AC: 99.7 P: 99.1 FM: 47.1	Higher processing speed Low training time	Low rate of detection of U2R and R2L attack types Low f-score. Dataset used does not reflect modern day attack types
[32]	2020	AE + PCA + GWO	NA	AC: 99.9 TNR: 99.9	High detection rate for binary classification Low computational complexity	No multiclass classification was performed, as a result could not show its capability at detecting various types of attacks
[33]	2021	IE-DBN + SMOTE + IG	KDDCUP99, NSL-KDD	AC: 98.1 AC: 98.7	An improved learning efficiency and detection accuracy leading to reduced computational cost and low false positive	Low detection rate of the minority class of attacks
[35]	2018	ABC AFS, CART FCM + CBFS	NSL-KDD UNSW-NB15	P: 99.0	High detection rate with low false positives	High overhead with respect to computational complexity and time cost. Low processing speed
[36]	2019	DNN LSTM-RNN DBN + DoublePSO	NSL-KDD CICIDS2017	AC: 99.7 P: 99.8 R: 99.8 FM: 99.8 FAR: 0.23 TNR: 99.7	High detection rate with significant reduction in the false positive when Double PSO was combined with DBN	Low detection rate of the underepresented classes due to class imbalance in dataset. Huge computational complexity

Table 4 (continued)

Paper	Year	Classifier	Dataset	Metrics	Pros	Cons
[27]	2021	DNN CNN XGBoost LSTM (LIO)	NSLKDD CIDDS001 CICIDS2017	AC: 87.0 P: 87.0 R: 83.0	Addressed class imbalance problem in data. Minimal expended time	Low detection rate
[17]	2020	Ensemble of Discriminant Approach	KDDCUP99	AC: 98.9 P: 99.7	Yield good detection accuracy for different types of attacks	Dataset used does not reflect contemporaries' attack. Also, class imbalance problem was not addressed
[37]	2018	SVM + GA	CICIDS20 ADFA-LDWMN	AC: 99.8 P: 99.4	Demonstrate the potential of detecting attacks with less computational complexity and less communication overhead	The dataset used to train the model only contains a small number of attack types, which could lead to misclassification when employed on networks with more attack types
[7]	2019	DT-SVM Ensemble	NSL-KDD	AC: 99.9 P: 99.9 FAR: 0.11	High detection rate	High False Positive
[41]	2020	M-AdaBoost SVM Ensemble + PSO		TPR: 99.0 AC: 99.0 P: 88.0 R: 95.0 FM: 91.0 FAR: 0.01	High detection accuracy Low false positive	High computational complexity
[40]	2020	PIO	KDDCUP99 NSL-KDD UNSW-NBS	TPR: 97.0 AC: 94.7 FM: 88.2 FAR: 9.7	Significant reduction of features to select most important subset leading to high detection accuracy, low false positive, and reduction in training time	Could not detect all types of attacks

Table 4 (continued)

Paper	Year	Classifier	Dataset	Metrics	Pros	Cons
[46]	2021	OC-SVM PIO	KDDCUP99 NSLKDD UNSWNB15	AC: 99.7 P: 99.8 FM: 99.2 FAR:0.01	Lightweight NIDS. High detection rate with low false positives	No multiclass classification was carried out to show the effectiveness of detection against different types of attacks
[60]	2020	SVM		AC: 96.2	Low false alarm rate	Low detection accuracy
[45]	2021	PABCE KSIDO	N/A	FNR: 90.0 P: 88.0	High detection accuracy for various attack types with minimized time and space complexity	There is no information on multiclass classification that demonstrate how well the model can recognize different types of attacks
[43]	2019	AMGA2 (MOGA)	NSLKDD	P: 88.0 FAR:0.02	Yield good accuracy in detection of anomalies	Requires iterative computations, and takes a lot of time, with high false positives
[44]	2021	O-CNN HM-LSTM LSO	NSLKDD ISCXIDS UNSWNB15	FNR: 5.7 AC: 96.3 P: 99.9 R: 95.8 FM: 98.1 FAR: 5.8	Effective performance in detecting anomalies	High computational complexity
[66]	2021	LNNLS-KH	NSL-KDD CICIDS2017	AC: 98.2 P: 97.7 FAR:1.18	An improved performance in accuracy and optimal fitness iteration curve and convergence speed with low false positive	Huge training time

Table 4 (continued)

Paper	Year	Classifier	Dataset	Metrics	Pros	Cons
[14]	2018	C4.5 BN FA	KDDCUP99	P:90.0 FM: 77.3 FAR:0.02	Indicated that 10 features is sufficient to build an improved classifier for IDS with high detection rate	High computational time
[10]	2019	MS-CNN + LSTM	UNSWB15	FNR: 1.6 AC:95.6 FAR:0.09	High overall detection accuracy with low false positive	High computational time, and Poor detection performance on imbalanced datasets
[38]	2020	AE	CSECICIDS2018	R: 99.7	Reduced computational complexity through feature selection and feature grouping. Addressed class imbalance problems	Low detection rate
[16]	2021	SVM + NB	UNSW-NB15 CICIDS2017 NSLKDD Kyoto 2006 +	AC: 93.7 P: 99.4 FAR: 3.0	High detection rate Low false positives	Class imbalance in dataset led to low detection of the minority class of the attacks
[8]	2021	TS-RF	UNSW-NB15	AC: 83.1	Reduced feature space/vector, therefore resulted in low computational complexity İmproved detection rate for low sample attacks Low false positive	Increased misclassification rate among different attacks types as a result of class imbalance problem in dataset
[13]	2021	DBN-SoftMax SDAE-ELM	KDDCUP99 ADFA NSLKDD UNSW-NB CIDDS-001	TPR: 0.04 AC: 78.0 P: 96.0 R: 64.0 FM: 76.0 FAR:0.06	High detection accuracy for binary and multiclass classification	Poor detection ability on small dataset High training time

Table 4 (continued)

Paper	Year	Classifier	Dataset	Metrics	Pros	Cons
[11]	2021	Light GBM Ensemble ADASYN Technique	NSLKDD UNSW-NB15 CICIDS2017	AC: 99.9 FAR:0.01	Application of ADASYN addressed the problem of imbalanced dataset, and led to an improved detection rate of the minority classes, with low computational complexity	False positive is relatively high
[39]	2020	Adaboost	KDDCUP99 UNSW-NB15 TRABID NSLKDD CICIDS2017	FAR: 0.02	Modest Adaboost has higher processing time but lower performance and higher error rate compared to Gentle and Real Adaboost	Gentle and Real Adaboost present good accuracy for binary classifications of intrusions
[56]	2019	PSO ACA GA	NSLKDD UNSW-NB15	TPR: 91.3 AC: 91.2 P: 91.6 FAR: 8.9	Enhanced specificity, precision and accuracy	Performance improvement for only binary classification
[50]	2021	DT SVM En-RF RFE	KDDCUP99	AC: 98.5 P: 95.0 R: 95.5 FM: 95.0	Good representation of dimensionality reduction leading to good classification rate for all the classes of attacks	Class imbalance in dataset led to low classification accuracy of the minority attack classes
[49]	2021	SAE + DNN	KDDCUP99 NSL-KDD UNSW-NB15	AC: 99.9 P: 99.9 R: 99.9 FM: 99.9 FAR:0.17	High detection accuracy rate	Fair representation of feature learning and dimensionality reduction
[48]	2021	GB-GWO	NSL-KDD	AC: 98.6	Enhanced accuracy towards attack detection	High time complexity

Table 4 (continued)

Paper	Year	Classifier	Dataset	Metrics	Pros	Cons
[47]	2022	DT,GBT,MLPAdaBoost,LSTM GRU	UNSWNB15 Network TON-IOT	AC: 99.0 R: 99.8 FM: 99.8	DT with the feature selection technique gave a promising result in detecting anomalies accurately when compared with other technique	The model did not extend its cability towards reflecting different types of attacks detected nor time expended
[28]	2020	CNN BG FCM,GA	NSLKDD	AC: 98.2 FAR: 0.5	Produced high quality feature set İmproved detection accuracy	High computational complexity İmbalanced data which led to biased classification towards the majority class
[67]	2021	RF		AC: 99.9 P: 99.9	High detection rate for binary classification	Could not effecively detect various types of attacks
[42]	2021	RCNN + LSTM	DARPA CSE-ICIDS2018	AC: 93.8 P: 99.9 R: 80.5 FM: 89.2	Detects various kinds of attacks with minimal error rate	High false positive rate, and low detection accuracy
[29]	2020	CNN	NSLKDD	AC: 94.0 P: 95.3	Detect anomalies of network traffic with huge semantic coding space	Low performance in anomaly detection for network traffic with low semantic space

TPR: True Positive Rate; FNR: False Negative Rate; AC: Accuracy; P: Precision; R: Recall; FM: F-Measure; FAR: False Alarm Rate; TNR: True Negative Rate

Table 5 Accuracy range for techniques

Study	No. of study	Techniques	Accuracy range (%)
[15, 27–30, 64]	6	CNN	(87–99)
[12, 31, 71]	3	ANN	(76.9–99.9)
[34, 71]	2	NB	(99.0–99.7)
[13, 33, 36]	3	DBN	(87.2–98.1)
[35]	1	ABC + AFS + CART	(99.0)
[36]	1	LSTM + RNN	(98.8)
[27]	1	XGBoost	(87.0)
[8, 54, 65, 67, 71]	6	RF	(83.1–99.9)
[27]	1	LSTM	(87.0)
[7, 37, 46, 60, 71]	5	SVM	(96.2–99.9)
[10, 42, 44, 69]	4	CNN + LSTM	(76.8–95.6)
[13, 32, 35]	3	AE	(87.2–99.7)
[16]	1	SVM + NB	(99.3)
[11]	1	Light GBM Ensemble	(99.9)
[39, 41, 52, 61, 71]	5	AdaBoost	(98.9–99.3)
[71]	1	KNN	(99.0)
[47, 50, 67]	3	DT	(85.2–99.9)
[49, 59]	2	AE + DNN	(89.0–99.9)
[53]	1	LR + SVM	(86.0)
[58]	1	XGBoost + DNN	(97.0)
[142]	1	CNN + AE	(100)
[62]	1	RNN	(94.5)
[68]	1	KELM	(95.8)

Table 6 The datasets used in the study

Datasets	No. of study
KDDCup99	19
NSL-KDD	25
CSECIC-IDS2018	4
CICIDS2017	8
ADFA-LD	2
UNSW-NB15	17
CICIDS-001	2
ISCX-IDS	1
Kyoto 2006 +	1
DARPA	1
RTNTP18	1
Network TON-IOT	1
UGR16	1

small number of samples for training [76]. However, it does not account for feature interdependencies while classifying data, which reduces the accuracy of the method [77].

K nearest neighbor (KNN) A supervised machine learning classifier known as KNN first determines the class of the k most similar trials before classifying the test sample. The sample test prediction is returned for the majority of these closest samples [78, 79]. These techniques aim to categorize an unlabeled data sample according to its k nearest neighbors. KNN does not need any parameters to function. The distance between neighbors is calculated using the Euclidean formula [78]. The fundamental idea behind the KNN classification algorithm is that fresh data instances are categorized into previously observed classes based on their relative proximity to each class. KNN-based classification has been employed by some studies [54, 71] for network intrusion detection with respectable accuracy in detecting attacks. Although KNN is easy to use, figuring out the ideal amount of k and locating missing nodes take time and are inaccurately expensive.

Decision tree (DT) In order to organize an ordered tree based on the value of a feature, decision trees (DTs) first extract features from the samples in a dataset. Each feature is represented by a node in the tree, and the branches arising from that node are used to represent the associated values of each characteristic. The origin node of the tree is any feature node that divides the tree into two parts in an optimal way [80].

Several criteria, including the Gini index [81] and information gain [82], are used to identify the origin node that best divides the training datasets. Induction and inference are two techniques used by DT algorithms to construct the model and then do the classification [83]. To determine the class of a packet, a trained DT chooses various aspects from the packet. The most data can be stored in an ideal DT with the fewest layers possible [82]. To create optimal trees, a number of techniques have been put forth, including Iterative Dichonomiser 3 (ID3), C4.5 [82], and classification and regression tree (CART) [84]. To evaluate DT performance, various measures are available. The use of ID3 is restricted by the fact that it does not support features with missing and continuous values and instead chooses characteristics with the maximum information gain (entropy). DT is a popular model in intrusion detection because it offers an easy-to-understand classification technique, is easy to understand and execute, and frequently enables superior generalization through post-construction pruning. DTs have the potential to be used as classifiers in the context of intrusion detection [85, 86]. However, factors like increased storage needs and computing complexity must be taken into account [83]. Another drawback of DT is its limited resilience, which means that even minor adjustments to the training data could produce a completely different DT. Larger DTs may need manual pruning because information gain is biased in favor of qualities with higher levels [87].

Support vector machine (SVM) Another kind of classifier is the SVM, which operates by constructing a hyperplane from the feature set of two or more classes. SVMs are especially suitable for the use case where classes with large feature sets need to be identified based on fewer data samples [88]. The splitting hyperplane is established by calculating the maximum distance of the nearest data point of each comparison class. There are numerous types of separating hyperplanes that can be accomplished using a kernel, such as a polynomial, linear, GRBF, or hyperbolic tangent. SVMs, which are based on statistical learning [89], are excellent for anomaly detection when it is necessary to classify data into normal and abnormal classifications. Due to their simplicity, SVMs are highly scalable and are able to carry out online learning as well as anomaly-based intrusion detection in real time [90, 91]. Utilizing SVM has the additional benefit of requiring less memory and storage. Several research papers [16, 37, 46, 60, 72] assessed the usage of SVM-based IDSs in network environments and found that SVM produced more accurate results than DTs, NB, and random forest. To separate the data when it is not linearly separable, SVM uses an ideal kernel function; however, this method still has limitations in terms of classification speed.

Random forest (RF) An example of a supervised ML algorithm is RFs. To forecast more accurate and error-resistant classification outcomes, an RF is constructed utilizing multiple DTs [92]. To output classification results based on majority vote, randomly built DTs are trained [92]. DTs and RF are both classification methods. However, DTs build a rule-set during training for the subsequent categorization of new samples, whereas RF builds a rule-subset using all member DTs. With far fewer inputs and without the need for feature selection, this produces a more reliable and accurate output that is resistant to overfitting [93]. According to several studies [8, 67, 71], RF is appropriate for network anomaly intrusion detection. Furthermore, another study [54, 94] found that RF outperformed KNN, ANN, and SVM in DDoS detection in networks because it used less input features and could avoid the time-consuming calculations needed for feature selection in real-time IDS[94].

Fuzzy C-means (FCM) The data are divided into clusters by FCM so that, in some cases, the data points with the greatest similarity belong to the same cluster and those with the greatest dissimilarity belong to distinct clusters [95]. Additionally, it offers additional conceptual advancements in a grouping and permits the assignment of a data point to many classes with variable degrees of association. As a result, it offers a more sensible method for handling patterns and has the ability to spot potential outliers [96].

Principal component analysis (PCA) PCA is a statistical method that has been widely used to identify prospective network traffic characteristics and eliminate superfluous or noisy resources. By removing the association between attributes and shortening training time, it aims to lower the dimensionality of the data [96]. Since its aim identifies the biggest variance with the smallest number of main components, PCA is a strategy that chooses a small number of uncorrelated features, known as main components, from a large number of features [97].

Extreme learning machine (ELM) The ELM is a single hidden layer feedforward neural network that selects the hidden-layer bias and input weights at random with no adjustment and calculates the output weights analytically [98]. In practice, this method often offers strong generalization performance at a very quick learning rate. To get beyond the drawbacks of ANN, a novel learning method (ELM) was presented by [68], the experimental findings support the suggested method's excellent accuracy as well as its time-saving advantages for network intrusions.

Convolutional neural network (CNN) The CNN is a shared-weight artificial neural network that is based on convolutional kernels or filters and was inspired by biological processes.

By utilizing equivariant representation, sparse interaction, and parameter sharing, it was created to reduce the amount of data inputs necessary for a traditional artificial neural network (ANN) [99]. The CNN is more scalable and takes less time to train and predict than the DNN due to its extraction of local features, which also reduces the number of weights and computational complexity. CNN is best recognized for its outstanding outcomes in computer vision and digital image processing, but more recently, it has also demonstrated promising results in its application in IDS. CNNs are often employed in the intrusion detection domain to extract features from the raw data. Convolutional layer, pooling layer, and activation unit are examples of three-layer kinds that are advantageous for CNN. Different kernels are used by the convolutional layers to convolute the data inputs [100]. By downsizing samples, the pooling layers reduce the sizes of subsequent layers. Max pooling and average pooling are the two methods used. Max pooling selects a maximum value for each cluster of previous layers after dividing the input into various clusters [101]. On the other hand, the average pooling determines the average values for each cluster in the previous layer. Every feature in the feature set can have an activation function triggered on it by the activation unit in a nonlinear way [101]. Although CNN excels in quickly and efficiently extracting features from raw data, it needs a lot of processing power [102].

Recurrent neural network (RNN) RNN is a DL method that works well in situations where data must be processed in order. Its output is dependent on back-propagation rather than forward propagation, unlike other neural networks [99, 103, 104]. A DNN has trouble fitting data that changes over time, hence the RNN was proposed as a solution. An RNN includes a temporal layer for sequential data analysis, followed by learning about multi-dimensional differences in hidden recurrent component units [105]. The neural network then makes adjustments to these hidden units in response to the data it encounters, resulting in ongoing updates and the manifestation of the neural network's current state. An RNN method processes the neural network's present unrevealed state by estimating upcoming hidden states as the triggering of a previously unrevealed state. Neurons of the previous layer receive feedback from neurons in the form of RNN function outputs. Although RNNs have been essential in fields like natural language processing and action recognition [106], they have recently become more pertinent and useful in IDS, where the majority of the data consists of temporally continuous data streams. An RNN has been proposed for use in prior research [36, 62] for network intrusion detection through study of network traffic behavior, and it has been stated that this method has produced useful results, particularly for time-series-based threats. RNNs can be utilized to effectively model the temporal correlations of malevolent behavior and security assaults. The probability of an attack is based on the present and prior states of the features, and the RNNs can be trained for this purpose utilizing both historical and current inputs [107]. The continuous product of their partial derivatives, however, can easily result in gradient disappearance or even gradient explosion when the number of layers in the network is significant because RNNs do not have a particular treatment for the activation function.

Long short-term memory (LSTM) The design of IDS has also utilized LSTM network designs, a particular type of RNN. By incorporating additional storage states, LSTM resolves the gradient vanishing issue that plagues the traditional RNN [108]. The ability to store data or cell state for subsequent use in the network is the primary characteristic of LSTM-based RNNs. They are suitable for performing analysis on temporal data that changes over time because of this property. LSTM uses a gate function as the activation function to selectively permit some information to pass through, which effectively controls the amount of gradient vanishing. Based on the original LSTM architecture, forgetting gates were added by [108] to allow the LSTM to reset its state and mimic memory forgetting. Because it resolves the gradient vanishing issue that plagues conventional RNNs, the LSTM has emerged as one of the most popular RNN types. Because LSTM networks are excellent for classification and prediction based on time-series data and because their forgetting mechanism is better suited for the detection of data streams, many IDS investigations use them [10, 42, 69]. However, the standard LSTM architecture cannot be trained in parallel due to the nature of RNNs [109], which makes LSTM-based models occasionally too expensive to use.

Autoencoder (AE) In feed-forward ANNs known as auto-encoders, the number of neurons in the input layer and output layer are equal. The inputs of an auto-encoder are constructed with the goal of minimizing the difference between the input and the output, and it is possible for it to contain numerous hidden layers. Each auto-encoder also includes a decoder and an encoder that map input data to the code and create input data from the code, respectively. By teaching the network to avoid signal noise, auto-encoders specifically support unsupervised learning of dataset encoding for dimensionality reduction [110]. The extraction of features from datasets is one use of AE. These, however, are hindered by the need for powerful computing.

Restricted Boltzmann machine (RBM) It is a method based on unsupervised learning that creates a deep generative and undirected model [111]. In any layer of an RBM, there are no two nodes that are connected in any way. The two kinds of layers that make up an RBM are visible layers and concealed

layers. The visible layer contains the known input parameters, whereas the hidden layer, which is made up of multiple layers, contains the unknown potential variables. Working hierarchically, latent variables representing features taken from a dataset are then passed on to the following layer. For network intrusion detection systems, RBMs were employed in a number of research studies [112, 113]. RBM implementation is difficult since it requires a lot of computer resources. Furthermore, feature representation is not possible with single RBM. However, by employing two or more RBM layered to create a deep belief network (DBN), this constraint can be solved.

Deep believe network (DBN) DBNs are generative probabilistic models made up of multiple stacked RBM modules. In a DBN, each RBM's output serves as the input for the one after it. In addition, neurons in the DBN layers have connections to neurons in the layer above them but not the other way around. DBNs can resolve ANN training issues and avoid issues such training that is too slow, requires a huge training dataset, or falls into a local minimum. DBNs are used in a variety of fields, including intrusion detection and has outperformed other ML algorithms in speech recognition, image identification, and natural language processing. These deep learning networks are effective in learning features and classifying data [114].

Ensemble learning (EL) To increase performance, ensemble learning combines several fundamental classifier models. The likelihood of choosing the incorrect classifier is decreased by an ensemble classifier's propensity for greater accuracy. You can stabilize the outcome [115] by pooling your guesses. By combining the findings of multiple classifiers and building on their advantages, EL creates a majority vote for categorization. Combining the output of different homogeneous/heterogeneous classifiers improves classification accuracy [116, 117]. EL is based on a study [118] that discovered that the accuracy of every ML classification algorithm depends on the application and related data. Bagging, Boosting, and Stacking are just a few ensemble strategies that can be applied. An aggregate classifier is produced using the bagging [119] method by using multiple iterations of the original classifier. As new learning sets for each version, bootstrap replicas of the learning set are created and used to create the various versions. The base classifier's instability is an important factor. Bagging can increase accuracy if altering the learning set can result in significant changes to the classifier that was built [119]. The main goal of the ensemble learning strategy known as "boosting" is to minimize bias and variation in supervised learning. To transform fundamental classifiers into an aggregate classifier is the goal. In contrast to bagging, the combination here is carried out by having each classifier operate inside the constraints of the prior classifiers. As a result, later basic classifiers are only tested or trained in scenarios where the earlier classifiers were unable to reach an acceptable level of precision [115]. The stacking technique takes into account a number of straightforward heterogeneous models, trains them concurrently, and then combines them to create a metamodel that makes predictions based on the predictions of several weak models. Since no ML technique can be said to be a "one size fits all solution," EL-like combinations may be best suited for generic applications to maximize accuracy through a reduction in variance and prevent overfitting [118]. Due to the use of many classifiers in parallel, the accuracy of EL comes at the expense of higher temporal complexity [120, 121]. Several research have looked at the effectiveness of EL for intrusion detection [7, 17, 63, 65]. According to research on the viability of EL under IDS, the EL algorithm provided more accurate and superior outcomes than each member classifier alone [122].

Extreme gradient boosting (XGBoost) XGBoost is an enhanced version of the gradient boosting algorithm [123]. In XGBoost, additive approaches are used to aggregate the predictions of weak learners to create a strong learner. It uses data from a previously cultivated tree to iteratively culture hundreds of trees. In order to increase accuracy in later iterations, this technique learns from the mistakes of earlier trees [124]. Because it employs a gradient-fall approach to reduce loss while introducing new models, the technique is known as gradient boost. Avoiding overfitting and making maximum use of computational resources are two additional advantages of XGBoost in addition to its advantages in speed and performance [124]. It employs the algorithms L1 (least absolute shrinkage and selection operator) and L2 to eliminate bias and overfitting.

Light gradient boosting machine (LightGBM) LightGBM is a gradient boosting framework that creates the final prediction by combining the findings of numerous DTs. It has a quick training time, uses less memory, and is dispersed and effective [125]. The gradient-based one-side sampling (GOSS) technique is used by the algorithm, in which sampling is carried out while keeping the instances with high gradients to preserve improved accuracy with the gain of information [125]. The strengths and weaknesses of the various strategies examined in the literature are displayed in Table 7.

4.4 Results Distribution of Signature-Based IDS

Signature-based IDS are rule-based and follow a process where signatures are searched and matched to detect intrusions. Pattern matching algorithms or rule-based techniques are often implemented for signature-based IDS. Evaluation

Table 7 Strength and limitations of ML- and DL-based methods

Paper	ML/DL methods	Pros	Cons
[16, 34, 71]	NB	Few samples are required for training. It can perform binary and multi-label classification. It exhibit robustness towards irrelevant features	interdependencies between features is not taken into account for classification purposes, therefore affects its accuracy
[54, 71]	KNN	It is Simple to use	It is quite challenging to determine the optimal value of K and identify missing nodes. Also, Prediction computation can be expensive
[14, 47, 50, 54, 63]	DT	It is simple and easy to use	Huge storage is required. High computational complexity. It is easier to use only when few DTs are used. Another difficult concern in DT is over-fitting
[7, 37, 46, 60, 72]	SVM	Highly scalable for its simplicity. Capable of detecting network anomalies in real-time, works well with learning online. It is suitable for data with large number of feature attributes. Lesser storage is required	The optimal kernel function used in separating data when it is not linearly separable remains an obstacle towards achieving desired classification speed. In the vicinity of the hyperplane, SVM are sensitive to noise
[8, 65, 67]	RF	It results in a more robust and accurate result that is resistant to overfitting. Fewer inputs are required and process of feature selection is not required	The use of RF may be impractical in real-time applications that require large dataset; this is because RF constructs several DTs
[13, 33, 36]	DBN	Suitable for vital feature extraction with training on unlabeled data	High computational costs
[13, 17, 41, 65, 68]	EL	It is robust to overfitting. Performs better than a single classifier. It reduces variance. Performs better in uncertain environments with a high number of features	Increased time complexity, due to the use of multiple classifiers in parallel
[36]	RNN	It is best-suited in environments where data are to be processed sequentially	The problem of vanishing gradients, which hinders learning of long data sequences
[15, 29, 30, 69, 142]	CNN	It is suitable for effective feature extraction from raw data. It possess good potential in network security application	High computational complexity, Consequently, it is difficult to implement them in a situation with limited resources
[38, 49, 59, 142]	AE	It is very effective for feature extraction and dimensionality reduction	It is computationally heavy. It may yield undesired results when the training dataset is not representative of the testing dataset

Table 7 (continued)

Paper	ML/DL methods	Pros	Cons
[33]	RBM	Feedback function of RBM facilitates the extraction of discriminative features from large and complex datasets which are then used to capture the behavior of the network traffic	Single RBM lacks the capability of feature representation High computational resources
[10, 27, 36, 42, 44]	LSTM	designed to address concerns with bursting or vanishing gradients	Training is expensive in terms of both time and resources
[13, 68]	ELM	Extremely quick rate of learning. For real-time retraining, they are intriguing	They are less reliable than conventional networks
[35]	FCM	It provides a more practical approach to addressing patterns and can identify potential outliers	The FCM can result in a local minimum
[32, 64]	PCA	Leads to a reduced training time	It is an attribute reduction technique that should be used in conjunction with additional prediction techniques to find incursions
[27, 58]	XGBoost	able to withstand overfitting, and full use of the computing resources	May struggle when dealing with outliers
[11]	LightGBM	Quick training and little memory utilization	Light GBM divides the tree along the leaves, which might cause overfitting because it results in trees that are more complicated. Since light GBM is susceptible to overfitting, it is likely to overfit little data

NB Naïve Bayes; *KNN* K nearest neighbor; *DT* decision tree; *RF* random forest; *FCM* Fuzzy C-Means; *SVM* support vector machine; *PCA* principal component analysis; *ELM* extreme learning machine; *CNN* convolutional neural network; *LSTM* long short-term memory; *RNN* recurrent neural network; *AE* autoencoder; *RBM* restricted Boltzmann machine; *DBN* deep believe network; *EL* ensemble learning

metrics frequently employed for signature-based are processing time, throughputs, memory usage, accuracy, precision, recall, F-measure, false-positive rate (FPR), which is sometimes called false alarm rate (FAR), true-positive rate (TPR) and missing rate (MR), also known as false-negative rate (FNR).

Standard benchmarks were proposed by [126] to establish common metrics that can be employed to evaluate how well different pattern matching algorithms works in detecting intrusions. Four algorithms, Morris–Pratt (KMP), Aho—Corasick (AC), Rabin–Karp (RK), and Knuth-Wu-Manber (WM), were implemented. The performance of Rabin–Karp with respect to memory and run time was found to be poor. Furthermore, in terms of memory consumed, the results revealed that KMP outperformed WM, while WM performed better than AC. For large datasets, KMP is more memory-efficient because it adds a second table to prevent recurring matches. In comparison with KMP, WM generates three extra hashtables and uses more memory; however, its run time is lesser. On the other hand, AC builds a large deterministic finite automata (DFA) trie, which explodes exponentially in size for larger signatures datasets. However, the algorithm was implemented in C#, which constitutes a garbage collection mechanism and makes memory measurements less reliable. In addition, it is good to note that the choice of programming language is an essential factor that can affect the

time and memory consumption of the technique. In general, more prolonged traces consume more memory for all three algorithms.

ManoharNaik and Geethanjali [127] proposed a multi-fusion pattern matching algorithm (MFPM) that incorporates the fusion of Raita and Berry–Ravindran algorithms. Berry–Ravindran with slight modification was implemented for the preprocessing phase which sets the pattern window's shift value (distance). This shift value is used during the searching phase to find patterns in the text using the fewest character comparisons feasible. Raita algorithms were implemented for the searching phase. The result of the study shows that the proposed MFPM algorithm has performed better than the existing algorithms in terms of the processing time (0.94) in milliseconds.

An IDS model based on combinatorial algorithms (CA) was proposed in [128]. To help the signature-based system (SBS) function as an anomaly-based system (ABS) and detect new attacks while also accelerating network traffic analysis by the combinatorial algorithm, the CA deployed extra databases. In order to increase the IDS's accuracy and reduce the false-positive rate, a suitable threshold of 12 was also established based on the analysis of prior research. The results indicated that CA had a higher accuracy of 96.5%, but a higher false-positive rate of 3%. Additionally, the dataset used is not clearly described, and there is a low detection rate for different types of attacks. Rao and Raju [129] proposed a MapReduce with pattern matching mechanism to accelerate signature-based intrusion detection. The experiment parallelized pattern matching technique under the MapReduce framework on CPU and GPU. The file size and searching time used by both CPU and GPU in the implemented pattern matching algorithm under MapReduce were compared. However, the model was only evaluated using run-time speed; other metrics were not considered, nor were the different types of attacks taken into consideration. The different investigations using signature-based are displayed in Table 8.

4.4.1 Pattern Matching/Rule-Based Algorithms for Signature-Based IDS

This session discussed a brief overview of algorithms and techniques that has been used for signature-based IDS. Table 9 presents the strength and limitations of each of the discussed algorithms.

Combinatorial algorithm The specific task of searching a database of sequences for a known sequence in bioinformatics gave rise to the concept of combinatorial pattern matching algorithm [128]. Algorithm 1 illustrates the fundamental combinatorial pattern matching algorithm. Two inputs are required by Algorithm 1 in order to produce the desired pattern. The two inputs are the network traffic t and the predetermined pattern p. Starting at point i the process will scan the substring of t with length n in ascending order of t_i=t_i... t_{i+n-1}. This algorithm supposedly exhibits the pattern in the following ways. İf t_i=p:

Algorithm 1Combinatorial pattern matching (p, t)
Input: pattern: p, text: t
Output: substring t matching p from position i
Initiate
n ← length of recognised pattern p
m ← length of traffic data t
for i ← 1 to m - n + 1
if t_i = p
output i
end if
end for

In the study by [128], the CA was employed to assist the analysis engine that was paired with the database of typical traffic in employing a score system to categorize the unknown traffic as either a match or a mismatch.

Aho–Corasick (AC) AC is a multiple matching method that constructs a deterministic finite automata (DFA) [130]. Using the "goto" and "failure" stages, AC encodes all patterns that need to be searched into DFA. One character at a time, AC creates a series of patterns in the initial "goto" step. The non-matching state is represented by the root node or state. For each character in each pattern in the dataset, AC creates a new node from the root node. For the remaining pattern characters, the procedure is repeated [126]. The depth of the state is determined by the longest pattern and is measured from the state to the root. The "failure" step deals with linkages that, in the event of a mismatch, only partially match another pattern. When the matching is unsuccessful, we return to the root node rather than the patching suffix with the largest length. Finding a matching pattern is done by simply traversing the trie while reading each character in the packet one at a time. Reaching the branch's end signifies that the matching pattern has been successfully matched; otherwise, no matches are registered [131]. AC performs better than other methods, particularly for shorter patterns, but as the number of patterns increases, so does the trie depth, increasing memory requirements dramatically. The trie needs to be reprogrammed each time a pattern is introduced.

Knuth–Morris–Pratt (KMP) KMP is the first algorithm for matching patterns in linear time. It is based on nave Bayes with fewer comparisons, [126]. If there are matches, KMP moves the pattern one position forward and repeats the

Table 8 Techniques, datasets, and performance of signature-based studies models

Paper	Year	Methodology	Dataset	Metrics	Pros	Cons
[143]	2021	FSM Based Regex Pattern Maching Algorithm	N/A	MU: 1.3 Mb TP: 593.4	Memory Efficiency and High throughput rate	No clear description on the types of attacks detected or dataset employed
[144]	2019	Similarity & IP Blacklist	N/A	AC: 99.4 PT: 25 s	Use of multiple smaller databases resulted in enhancement with respect to time spent to detect the attacks. Also, the use of similarity and an IP blacklist facilitated the detection of new attack signature	No adequate information on datasets used, neither was there a clear description on the attack types been detected. Also, the memory complexity
[128]	2016	Combinatorial Algorithm	N/A	TPR: 96.0 AC: 96.5 FAR: 3.0	High detection rate	High false positive rate, and did not cover wide range of attacks
[70]	2019	DT Models (C5, CHAID, CART, QUEST)	UNSWB15 RTNTP18	AC: 83.8 DR: 90.3 FM: 84.5 FAR: 2.0	High detection rate with low false positive	Not able to detect zero-day attacks, and not able to detect on real-time data set
[145]	2017	Parallel Processing with Snort Rules	N/A	PT: 7m31s	Using multithreaded technique, the problem of database size and infrequent signatures were resolved with decreased matching time	The process of selecting signatures is manual, which makes it difficult to update the database of signatures
[129]	2019	MapReduce with Pattern Matching Mechanism	N/A	MU: 10gb PT: 26 m	High execution speed. Reduced searching time	Focused only on few types of attacks
[126]	2018	AC, KMP, WM, RK	N/A	MU: 740,792 PT:0.03 m	KMP consumes less memory as it creates additional tables to avoid repeating matches. WM use less run-time	Generally space and time complexity needs to be improved for matching patterns
[127]	2016	RA, BRA	N/A	PT: 0.94 ms	Low processing time	Low detection rate of various kinds of attacks

TPR True Positive Rate; *MU* Memory Used; *AC* Accuracy; *DR* Detection Rate; *FM* F-Measure; *FAR* False Alarm Rate; *PT* Processing Time; *TP* Throughput; *NA* Not Available

matching process, otherwise it compares each character of the pattern with the packet. Pre-computed tables are used to reduce the need for repetitive matching. KMP performs exceptionally well for lengthy packet traces because, it does not need to match backwards into the packet like its predecessor. [132].

Wu-Manber (WM) The WM is an algorithm for multiple pattern matching that incorporates a B character sized sliding window [133] in addition to Boyer-poor Moore's character heuristic [134]. Shift, hash, and prefix tables are built by the algorithm to speed up the matching process. For various windows hashes, the shift table calculates the skip forward values. The shift table's associated value is read by WM after the last B characters of the sliding window are hashed. A shift of zero denotes a likely match, necessitating the algorithm's use of the prefix and hash tables to locate and verify the match. The WM algorithm performs poorly for shorter patterns, but it is quite effective for longer patterns [126].

Boyer–Moore algorithm (BM) Two tables, referred to as the Boyer–Moore bad character (bmBc) and Boyer–Moore good-suffix (bmGs) tables, are produced by the algorithm after preprocessing the pattern [127]. A bad-character table keeps the shift value for each character in the alphabet set based on how often that character appears in the pattern. A good-suffix table, on the other hand, keeps track of the matching shift value for each character in the pattern. During the searching phase, the greatest shift value between the bmBc (character in the text where a mismatch occurred) dependent expression and from the bmGs table for a matching suffix is

Table 9 Strength and limitations of pattern matching techniques

Paper	Algorithms	Pros	Cons
[128]	CA	Effective at classifying unknown traffic as either match or mismatch	High time complexity
[126]	AC	Good performance for shorter patterns matching	Due to the trie depth, there is exponential growth in the required memory with the number of patterns
[126]	KMP	KMP performs very well for large packet traces as it does not require a backward match in the packets	It exhibits longer run time
[126]	WM	WM is very efficient for larger patterns	WM performance degrades for shorter patterns
[127]	RA	Decreased Searching Time	High Resources constraints, Low detection rate for various types of attacks
[127]	BR	Quick Search and Effective in preprocessing of data packets	High Space Complexity for larger packets

CA Combinatorial Algorithm; *AC* Aho-Corasick; *KMP* Knuth-Morris-Pratt; *RA* Raita Algorithm; *WM* Wu-Manber; *BR* Berry-Ravindran Algorithm; *BM* Boyer-Moore Algorithm

taken into account after each try. Numerous pattern-matching algorithms are built on this approach [134].

Raita algorithm (RA) By comparing each character of a pattern in the given text, the Raita algorithm looks for that pattern. The following will be the search process. The rightmost character of the window is compared to the first and last character of the pattern. If a match is found, the first character of the pattern is compared to the character that is furthest to the left in the window. If they match once more, it compares the window's middle character to the pattern's middle character. If everything goes according to plan, the initial comparison begins with the second character and ends with the last character. The algorithm executes the bad character shift function that was calculated during the preprocessing phase if there is a mismatch at any point in the process. The bad character shift function is comparable to the Boyer–Moore algorithm's proposal [127].

Berry–Ravindran algorithm (BR) The Berry–Ravindran algorithm is a combination of the quick search method and the Zhu–Takaoka algorithms, a different variation of the Boyer–Moore algorithm [127]. By taking into account the "bad-character shift" for the two consecutive text characters to the right of the window, it conducts the window shifts. The preprocessing stage's computation of a two-dimensional array yields the shift values. The shift value of the pattern on text is only computed using this approach when two consecutive characters are used. During the searching phase, the greatest shift value between the bmBc (character in the text where a mismatch occurred) dependent expression and from the bmGs table for a matching suffix is taken into account after each try. Numerous pattern-matching algorithms are built on this approach. Berry–Ravindran and Raita algorithms have been combined in the [127] suggested algorithm. In the preprocessing step, the Berry–Ravindran bad character (brBc) function was shown to be efficient in setting the pattern window's shift value (distance) which is used during the searching phase to find patterns in the text using the fewest character comparisons feasible. Table 9 shows the analysis of the strength and limitations of the various algorithms used for the signature-based IDS in the reviewed literatures.

4.5 Result distribution of hybrid-based IDS

Hybrid-based combine the methodologies of anomaly-based and signature-based. Although signature-based IDS has a low FAR and is very effective at identifying known attacks, it struggles to identify novel or unidentified threats. Anomaly-based IDSs, in contrast, have a high FAR but a high detection rate of new and undiscovered assaults. Therefore, employing a hybrid approach seeks to balance out each strategy's strengths and weaknesses to produce greater performance at identifying network intrusions.

In adopting a hybrid-based approach, [135] suggested a paradigm for implementing a hybrid-based strategy called D-Sign, which is made up of three tiers of components: a honeypot server, a detection engine (DE), and a signature generation engine (SGE). On the NSL-KDD and CICIDS 2017 datasets, D-Sign was assessed. In order to redirect unusual traffic to these phony servers rather than the network's actual production servers, honeypot servers imitate vulnerable servers. To learn more about the attacks, DE further studied the traffic it had gathered from the honeypot's logs. To find attacks or anomalies in the packets kept by the honeypot server, DE employed a hybrid strategy combining the misuse detection engine (MDE) and anomaly detection engine (ADE). A rule-based approach was employed for the MDE and an LSTM-based recurrent neural network was used for the ADE to classify the attacks. If there is a match between signature patterns, MDE will sound an alarm. The data stream

is routed to ADE to find new attacks if there is no signature match. If DE identifies an attack, it sends the information to SGE, which creates a set of signatures for malicious traffic packets and keeps them in the IDS's signature repository. The model achieved a TPR of 99.45% and 98.51%, TNR of 99.32% and 99.20%, and accuracy of 99.10% and 99.40% for CICIDS2017 and NSL-KDD datasets, respectively. However, the system did not consider multiclass classification, the problem of class imbalanced dataset was not addressed, and the signature matching algorithm can be improved to increase the quality of signatures, according to the researchers. Additionally, metrics like FAR and processing time were not considered in the evaluation of the system.

Abbasi et al. [73] presented a hybrid technique based on optimization of pattern matching (OPM) and deep learning-based feature extraction (DLFE) for NIDS. Features were extracted using AE, attacks were categorized using SVM, and patterns were matched using Snort rules. According to the outcome, the approach performed well in terms of time, throughput, and memory. According to the study, the OPM phase was evaluated using packet processing time, throughput, and memory utilization, while the DLFE phase was examined using performance characteristics such as accuracy, precision, and FAR. The length of time it takes for a pattern matching engine to process a data packet increases as it gets bigger. The Snort pattern matching engine was outperformed by the proposed model in terms of time complexity. Other attack kinds were not taken into account by the model, though. However, there is not enough data available regarding the outcomes of the DLFE phase. Additionally, the study stated that each phase was examined independently, although there is little information on how the various modules interact.

Kim et al. [85] offered another hybrid-based IDS, and the NSL-KDD data set was subjected to decision tree and 1-class SVM analysis. The C4.5 decision tree (DT), which was also utilized to split up the regular training data into smaller subsets, was used to develop the misuse detection model. Then, a model for detecting anomalies was built utilizing a single-class SVM in each deconstructed region. This model can indirectly leverage knowledge of past attacks to improve its performance when creating profiles of typical behavior. The experimental findings showed that the proposed model might enhance the IDS's performance and speed in identifying unknown threats while also reducing the training and testing procedures' time complexity. The model's overall average training and testing times were 56.58 and 11.20 s, respectively. However, because the original C4.5 decision tree does not take into account clusters in the normal data set, it can split a well-formed normal cluster during the decomposition processes. This can impede the well-formed normal cluster that is associated with a single decision boundary in the 1-class SVM model and reduce the model's capacity for profiling. Additionally, when there is an unequal distribution of data, training and testing take longer. As a result, the suggested method's capacity for time reduction falls short of expectations. In order to enhance the model's performance without reducing its capacity to identify unidentified attacks, different classifiers must be investigated.

In the same vein, using a greedy algorithm-based global optimal Gaussian mixture model (GA-GOGMM-based) and fuzzy rough set-based feature selection (FRS-FS) pattern learning, [136] suggested an adaptive network intrusion detection (ANID) technique. By calculating the information gain ratio (IGR) of each potentially sensitive characteristic based on the FRS theory, FRS-FS determines the best attribute subset. GA-GOGM was used to detect the number of clusters to automatically extract the pattern of both normal and attack network connection instances for NIDS based on an incremental learning method. The normal and attack pattern libraries were updated at the same time using an online-based updating process that mined popular patterns and eliminated out-of-date ones. As a result, the suggested NID model had low false positive and missed reporting rates while achieving high accuracy for known and unknown attacks. The model had an overall identification rate of 96.52% when it was tested using NSL-KDD and a self-built Nidsbench-based dataset. The model's FAR for known attacks was 1.43%, MR of 2.56%, FM of 97.45%, for unknown attacks, a FAR of 0.25%, MR of 15%, and FM of 87.67% was achieved.

Similar to this, a genetic algorithm was employed as a filter strategy for choosing the quantity of features from a given dataset in the hybrid model presented by [137]. With the help of these chosen features, the model was trained using the C4.5 decision tree (J48) as a classifier, and after that, association rule mining was utilized to produce rules for the creation of a normal traffic rule database for the intrusion detection model. The findings show that the combined impact of both signature- and anomaly-based systems improves IDS performance. KDDCup99, DARPA Lincoln, NSL-KDD, Kyoto, and UGR16 datasets were used to evaluate the model. The experimental findings demonstrate that the systems attained a 99.96%, 99.81%, 99.89%, 99.94% and 99.32%, accuracy rate. A DR of 99.96%, 99.78%, 99.88%, 99.32%, 99.99%, and FAR of 0.01, 0.23, 0.01, 0.00, 0.2 for KDDCup99, DARPA NSL-KDD, Kyoto 2006+ and UGR16 dataset, respectively, although the model did not consider multiclass classification.

138 Proposed a hybrid model that combined misuse and anomaly approach for anomaly detection in networks using Snot and naïve Bayes. Snort was applied for known attack detection and result in low false positive. Feature selection technique was introduced to select 25 significant features, which then serve as an input to the naive Bayes classifier.

Table 10 Techniques, datasets, and performance of hybrid-based studies models

Paper	Year	Methodology	Dataset	Metrics	Pros	Cons
[9]	2020	RNN LSTM Bro Signature Analyzer	N/A	TPR: 96.4 FNR: 4.1 FAR: 2.7	Good detection rate for zero-day attacks	High false positive
[136]	2020	FRS-FS GA-GOGMM	NSL-KDD, Nidsbench-based	DR: 96.5 FAR: 1.4	Low false positive and missing rate	Low accuracy for known and unknown attacks
[138]	2020	Snort + Naïve Bayes			An increased execution speed with decreased processing time	Low detection accuracy rate, and huge resource consumed
[135]	2019	rule-based approach and RNN using LSTM	CICIDS 2017, NSL-KDD	TPR: 99.4 AC: 99.1 TNR: 99.2	High detection accuracy rate	Time expended by the model is relatively high, also imbalance nature of data leads to low detection of some specific kinds of attacks
[137]	2019	DT, Association rule mining, GA	DARPA,KDDCup99, NSL-KDD, UGR16 Kyoto 2006 +	AC:99.9 DR: 99.8 FAR:0.2	Enhanced detection performance	Different attack types were not explored
[73]	2021	AE, SVM Snort AC Rules Algorithm	NA	PT:1.6us/100byte TP:621,118/100byte MU:682.7 Mb/5byte	An improved time complexity to process each packet	Only detects few types of attacks
[85]	2014	C4.5 decision tree (DT), 1-class SVM	NSL-KDD	PT: 11.2 s	High detection performance for unknown attacks, low processing time, and less training time	Degraded profiling ability Uneven data distribution leading to high training and testing time

Result showed an increased accuracy of 97.5% for the proposed hybrid model, and also its capability to detect new type of attacks. With an increased execution speed and decreased processing time. In another study, [9] proposed an IDS based on recurrent neural networks (RNNs), which is applied to find complex patterns in attacks and generate similar ones. The authors demonstrated that RNNs can be very helpful at generating new types of attacks which can sum up to form a dataset of unseen attacks signatures. The method achieved good detection rate for zero-day attacks as it improves the ability of NIDS to detect new attack signature it has never seen before, although with high false positive. Table 10 shows the related work on the hybrid-based approach.

5 Summary of discussion

In responding to RQ1, based on the analyzed result, various machine/deep learning techniques (CNN, ANN, DBN, RF, AdaBoost, LSTM, SVM, RNN, EL, AE, DT) have been independently employed by some research for anomaly-based IDS, while some researches have employed a hybrid of two or more techniques (CNN+AE, AE+DNN, CNN+LSTM, LR+SVM, XGBoost+DNN) to improve performance. We can observe that conventional machine learning techniques are still widely used. They are more interpretable, less reliant on processing power, and easier to deploy and implement than deep learning techniques. However, it was evident that there has been a shift to deep learning (DL) techniques, which is now the current trend as illustrated in Tables 4 and 5 which indicated that majority of the literatures applied deep learning techniques, with CNN been the most applied. The rapid development of deep learning has contributed to research of intrusion detection. Supervised learning is mostly preferred by researchers, which may be because many publicly available datasets are already labeled, however in the absence of labelled dataset, annotation is time-consuming and labor-intensive especially for large datasets, additionally at the time of practical application, data obtained will be unlabeled, therefore it becomes imperative to consider un-supervised and semi-supervised learning as they will be the way forward for network anomaly detection. For the signature-based IDS, pattern matching/rule-based

approach (*Aho–Corasick, combinatorial algorithm, Knuth—Morris–Pratt, Wu–Manber, Boyer–Moore Algorithm)* are the most commonly employed algorithm in the literature. In the same vein, the Hybrid-Based IDS explored in the literature shows a combination of signature + anomaly-based techniques such as RNN LSTM + bro signature analyzer; Snort + naïve Bayes; AE, SVM + Snort AC rules algorithm.

In responding to RQ5, with respect to the performance level of the various techniques proposed for the anomaly-based IDS, the result of the analysis revealed that hybrid of CNN and AE gives close to 100% performance as indicated in Tables 4 and 5. AE techniques have the ability to detect anomalies by taking advantage of the nonlinear and linear dimensionality reduction, AE also have the ability to deal with high dimensional, unstructured, and heterogeneity data generated from network traffics. İn addition, the training phase of AE involves the regeneration of new input data from partially destroyed one. Automatic detection of important feature as well as independent learning of the key feature of each class is one of the attributes of CNN, which is why it is considered to be a vast technique. Nonetheless, some factors like kernel size, padding, filter, and stride may affect the efficiency of CNN, which explains why the combination of AE with CNN may likely reduce the high dimensional data, thereby leading to a promising result of recall, precision, and accuracy. Although, the computational cost, training time, resource consumption and the types of attacks to be detected is worth noting. For the signature-based IDS, various algorithms such as Aho–Corasick algorithm (AC), Knuth–Morris–Pratt (KMP), Wu–Manber (WM), combinatorial algorithm, Berry–Ravindran and Raita have been explored in the literature proposed to speed up the process of matching signatures, however, it is challenging to make a comparison of the performance of these approaches due to the different metrics, datasets, and simulation environments that were employed. Nonetheless, due to the operation of the pattern matching or signature search, existing techniques bear more memory and searching time. Therefore, this calls for a diligent system that can reduce the overhead. On the part of the hybrid-based IDS explored in the literature, few studies were found in this regard, whose methods indicated a good detection rate but with a high false alarm rate, also the time expended by these models are relatively high.

In responding to RQ3, Table 6 and Fig. 4 depict that anomaly-based IDS are evaluated using metrics such as accuracy, precision, recall, F-measure, false alarm rate (FAR), false negative rate (FNR), true positive rate (TPR), and true negative rate (TNR), among which accuracy precision, recall, F-measure, and FAR are the most commonly used metrics. The study further established the importance of the FAR metrics also presented as false positives, whose large numbers will put more strain on the system and human resources. FNR has also been identify as key indicator of the model effectiveness, and a good IDS model should take into account this crucial metrics because if a model is said to have high FNR, that can translate into lack of sufficient protection for the system making it more vulnerable to attacks, Additionally, Precision metrics has been illustrated as a good performance indicator which may translate into reduced misclassification rate for the attacks. For signature-based IDS, the result outlined memory utilization, processing time, accuracy, as the metrics often used as performance indicator of the models, these metrics are deemed significant due to the extensive signature database that has to be sustained, updated, and compared with the data packets for possible attacks. For the hybrid-based IDS, it takes both metrics of the anomaly-based and signature-based into account, however, from the findings of the analysis conducted, it appears that, the performance evaluation of the hybrid-based IDS is often independent of the signature-based and anomaly-based, there is need to find a common ground whereby the entire system can be evaluated. Another typical evaluation parameter used in IDS methods is detection time. It calculates how long it takes to categorize a sample using a trained model. Even with techniques like feature selection for dimensionality reduction, IDS research frequently encounters the issue of dimensional disaster due to the complexity of network traffic, which eventually manifests as a long detection time. The high detection times of some of the many known intrusion detection algorithms make them practically impossible to use in real-world applications. A model's long detection time typically indicates that the algorithm complexity is too high. Studies that have been looked into have demonstrated a definite tradeoff between performance and complexity models. Even though computational complexity has a direct impact on detection time, most papers only report the training and testing times of their algorithms on the given dataset because it can it can be challenging or controversial to determine the computational complexity of some algorithms under various assumptions. It is still challenging to assess an algorithm's superiority in terms of time complexity solely by looking at its running time because the platforms utilized to acquire each result and the dataset preprocessing techniques vary. In conclusion, we think that rather than merely looking at detection time, there is still a need for a unified complexity evaluation standard in the current IDS research.

In responding to RQ2 and RQ4, the result of the analysis demonstrates that NSL-KDD and KDDCup99 are the most often used datasets for anomaly-based IDS, both of which are simulation experimental data. This is primarily due to the fact that KDD99 and NSL-KDD are datasets that have long been made available to the general public. On the basis of these two datasets, many articles by researchers have been published. The constant use of KDD99 and NSL-KDD results from the necessity of comparing new intrusion

detection systems with existing ones. However, the contents of these datasets are obsolete, and do not represent contemporary attack types. Among the Attacks are in the categories of user to root (U2R), remote to local (R2L), denial of service (DoS), and probing were identified by some model. Similarly, in some instance, CICIDS2017, CSE-CIC-IDS2018 were used which tackled few contemporary attacks such as Shellcode, Fuzzer and worm, exploits, back-doors, analysis, generic, and reconnaissance Bruteforce, Heartbled, Bot, Web attack, DoS-Hulk, Infilteration, DoS-GoldenEye, DoS-lowHTTPTest, DDoS-LOIC-HTTP, DoSSlowloris, and DDoS-HOIC. A variety of attacks, including DoS/DDoS, Botnet, Web Attack, PortScan, Bot, grey hole, black hole, SQL injection, Benign, DoS Hulk, FTP-Potato, SSH-Potato, Dos Slowhttptest. The types of attacks an IDS is capable of detecting is an essential factor in identifying an effective IDS. While some of the models in the study have achieved great accuracy in identifying a certain attack type, their studies didn't address modern types of attacks or zero-day attacks. Therefore, there is a need for a model that can detect various attacks with higher detection rate.

6 Challenges and potential future directions

Network IDSs have been the subject of numerous studies and research papers that have been published. In response to RQ 6, the use of ML and DL algorithms for anomaly-based network intrusion detection systems, in particular, still faces several open research hurdles and problems. The problem is that there isn't a common mechanism that ensures the proposed systems' or method's validity. The majority of research studies provide evaluation of the suggested systems using simulated datasets, which may not be applicable to real-world situations with real data and other challenges. As demonstrated by this study and others like it on the state of the art in IDS, it is highly challenging to build an IDS which encompasses, at least, the most important aspects of an effective IDS, namely, It can be quickly deployed online, is scalable, effective with real data, and meets the requirements of all stakeholders. Instead, the majority of the published work presents results using biased parameters, covers only a small portion of the system, and shares evaluation results tested on fabricated datasets.

In the context of signature-based IDS, a variety of techniques, such as pattern matching and rule-based algorithms, have been documented in the literature with the primary focus on accelerating the process of matching signatures (time complexity), while researchers paid little attention to space complexity (memory consumption), especially when there is an increasing size in database. Due to the many measurements, datasets, and simulation environments, it is difficult to compare the performances of these methodologies. The pattern matching or signature search processes, however, need greater memory and search time in current techniques. As a result, this necessitates an enhanced strategy that can optimize memory usage and processing time.

For hybrid-based IDSs explored in the literature, only seven studies [9, 73, 85, 135–138] were found in this regard. The studies under review didn't provide sufficient information on the attacks detected as multiclass classification was not employed. In addition, there is a lack of sufficient background information on how the whole system has been integrated and evaluated.

Furthermore, it is extremely difficult to define or achieve a demonstration of the accuracy and completeness of any proposed IDS. As a result, this study's key finding is that it is challenging to create an all-encompassing IDS that provides high levels of accuracy, scalability, resilience, and threat prevention. The main problems and difficulties that scholars deal with now and in the future are discussed below. Future research in this field has a huge potential, especially in hybrid-based intrusion detection that uses Signature and Anomaly approaches. Although there has been a lot of study done in the IDS setting, the following factors of the most recent Network IDS issues can be taken into account in future studies:

- A high-quality dataset is absolutely necessary to test and validate proposed NIDS. Such a dataset should include a sizable amount of network traffic data with labels that describe both attack and regular behavior. We advise that future investigations assess the effectiveness of intrusion detection techniques utilizing more recent intrusion detection datasets, such as CIRA-CIC-DoHBrw 2020 dataset. In order to confirm the viability of their strategy, researchers should also try to experiment with some genuine datasets, such the ISOT CID dataset. In addition, researchers should try to experiment with some real datasets, such as the ISOT CID dataset, to ensure the validity of their approach.
- It is extremely difficult to develop an online, real-time network IDS. This is due to the fact that an IDS of this type would need to first understand regular behavior in order to detect anomalous or malicious behavior. The learning phase presupposes the absence of noise and attack traffic, which cannot be guaranteed. If these problems are not resolved, such an IDS can cause false alarms.
- The majority of anomaly-based NIDS attempt to build a model that captures the profile of every potential action or pattern of regular traffic. This is particularly difficult, though, as it has been demonstrated that these models have a tendency to favor the dominating class, leading to large false-positive rates. Additionally, it is impossible to catch every single potential normal observation that might be

produced in a network, which raises the rate of false-negative results. Another research goal is to completely prevent or reduce false-positive and false-negative rates in NIDS.

- During various phases of the design and implementation of NIDS, computational complexity increases, such as feature reduction and data preprocessing, model training, and deployment, in particular, ML- and DL-based NIDS. Consequently, creating an effective NIDS with minimal computational requirements is a challenge and a potential field for future research.
- The feature selection and dimensionality reduction techniques employed in the proposed IDSs are appropriate for a particular sort of normal traffic and a specific type of attack detection, but they could not be effective if the environment for normal or attack sequences changes somewhat. Therefore, a major research goal is to develop a dynamic and computationally efficient technique for feature selection that can operate in all kinds of normal and attack traffic.
- For the purpose of training a model on a sizable dataset, DL and ML approach are frequently employed. This has made it easier to handle cyberattacks effectively. However, some issues regarding the application of DL and ML approach for attack detection in networks require the attention of researchers; for instance, the issue of resource limitations limits the application of DL/ML algorithms [43, 63] for network security [139]. Scalability problems, such as those related to various scenarios and IDS deployment options, are another difficulty with the use of ML/DL techniques in large, distributed networks. One possible solution to limitations of individual deep or machine learning methods as recommended by some of the authors [63] is the usage of an ensemble of ML/DL algorithms that outperform a single machine learning algorithm. However, because these algorithms were computationally expensive, they caused network latency problems, which are unaffordable in critical network environments.
- The methods of semi-supervised learning, transfer learning, and reinforcement learning (RL) have not yet been thoroughly investigated and tested for creating an IDS for network security. Therefore, future research may focus on the semi-supervised learning, transfer learning, and reinforcement learning (RL) techniques for creating an IDS for network security so as to accomplish crucial goals like real-time, quick training, and unified models for anomaly detection [139]. Additionally, combining RL with DL would make for an intriguing research topic because their combined use may be advantageous in network systems with enormous data dimensionality and non-stationary situations.
- Finally, it would be interesting if more studies can focus on development of a hybrid system that combines techniques of signature and anomaly-based for IDS, as it is evident from literature that this aspect has not been thoroughly investigated and tested for creating IDS.

7 Conclusion

In this work, a survey of anomaly-, signature-, and hybrid-based IDS has been presented. This work attempts to provide the researchers with the summarized but comprehensive and useful insight into the various security challenges currently being faced and possible solutions, with a focus on network intrusion detection.

This study adheres to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) standards. Literature on signature-based, anomaly-based, and hybrid-based network intrusion detection was organized and summarized according to a set of predefined keywords and RQs. A total of 71 articles were chosen for the review after the exclusion, inclusion, and quality criteria were applied. This study gave insight into what may be learned from existing research on hybrid-, anomaly-, and signature-based network intrusion detection. Additionally, it compared methodology, datasets, types of attacks detected, evaluation metrics, performance and limitations of each contribution.

The analysis of the SLR has shown that there has been a tremendous shift from signature-based towards the anomaly-based approach in terms of research studies; however, the field of hybrid-based approach has minimal focus by researchers. The study further shows that different machine learning and deep learning algorithms have been deployed by researchers for intrusion detection, among which the commonly used ones include CNN, ANN, DBN, AE, AdaBoost, LSTM, RF, and SVM, while in terms of accuracy, CNN+AE was reported to yield better performance.

In the analyzed studies, thirteen types of datasets were used for evaluating the models. KDDCup99 and NSL-KDD were the most used dataset, followed by the UNSWNB15. These datasets have been criticized by researchers for being out of date because they do not reflect the contemporary sorts of assaults, despite their high recorded popularity. To this end, the SLR analysis revealed emerging datasets that contain modern day attacks. These include CIRA-CIC-DoHBrw 2020, ISOT CID, CICIDS2017, ADFA-LD, ISCX-IDS, RTNTP18, UGR16, and CSE-CIC-IDS2018 datasets. Therefore, it is important for researchers to utilize modern datasets in order to attain an effective IDS.

For anomaly-based IDS, it was evident that deep learning (DL) techniques are now the current trend. Majority of the reviewed literature applied deep learning techniques. In terms of performance, there was report of nearly 100% performance, based on CNN and AE. However, when considering the various attack types found in the datasets, it is essential to note that the FAR needs to be reduced to the minimum. We

suggest that studies explore other datasets with more recent types of attacks. Additionally, some DL techniques are yet to be explored, which gives room for further research to be done in this area. A hybrid of two or more techniques may yield high detection accuracy for attacks but may result in high training time and resource consumption. Therefore, we recommend that future research look into how a practical, lightweight anomaly-based IDS should be modelled, considering all the challenges mentioned above.

Several methods have been proposed to speed up the process of matching signatures. In the context of signature-based IDS, pattern matching and rule-based algorithms are the most used in the literature. However, it is challenging to compare the performances of these approaches due to the different metrics, datasets, and simulation environments. Nonetheless, due to the pattern matching or signature search operations, existing techniques bear more memory and searching time. Therefore, this calls for an improved approach that is capable of optimizing processing time and memory consumption.

For hybrid-based IDSs explored in the literature, only seven studies were found in this regard. The studies under review did not provide sufficient information on the attacks detected as multiclass classification was not employed. In addition, there is a lack of sufficient background information on how the whole system has been evaluated. Therefore, there is a need to design a hybrid system that overcomes the aforementioned reported issues.

Further analysis of the review reveals that while processing time and memory usage were employed for signature-based IDS performance evaluation, accuracy, precision, and FAR were the three most often used metrics for anomaly-based IDS. Future research should think about including all different performance indicators, especially detection time, which will be crucial for real-time systems. IDS models should also be able to adapt to the dynamic nature of the network environment; this calls for models that can autonomously learn their surroundings and update themselves.

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